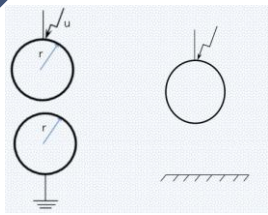
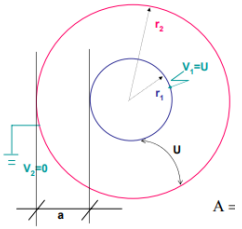
EE4047 – High Voltage Techniques I

Continuation to:
Cylindrical Electrode Systems





Week 10.1 – Monday Lesson

Asst. Prof. Dr. M. Alparslan Zehir

08.05.2023

OUTLINE

2 Multilayer Dielectric Concentric Spheres

3 MULTILAYER DIELECTRIC COAXIAL CYLINDERS

Multilayer dielectric materials between the electrodes:

$$I = I_1 = I_2 = I_3 = \cdots = I_n$$

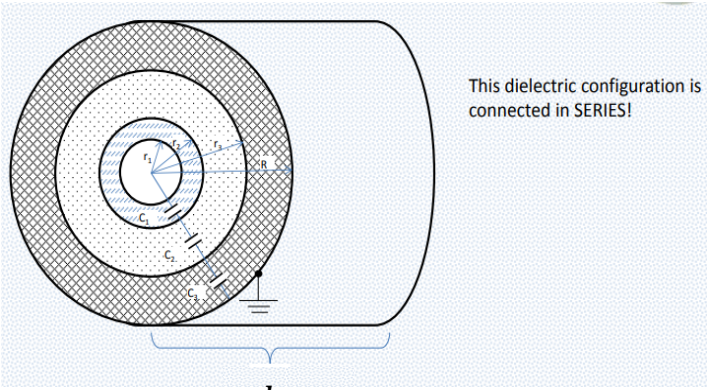
$$Q = Q_1 = Q_2 = Q_3 = \cdots = Q_n$$

$$C\,U = C_1U_1 = C_2U_2 = C_3U_3 = \cdots = C_nU_n$$

$$U = U_1 + U_2 + U_3 + \cdots + U_n$$

U is the voltage applied to the entire system

U_1, U_2, …, U_n are voltages across the corresponding dielectric layers



3 MULTILAYER DIELECTRIC COAXIAL CYLINDERS

Voltages across each dielectric layer:

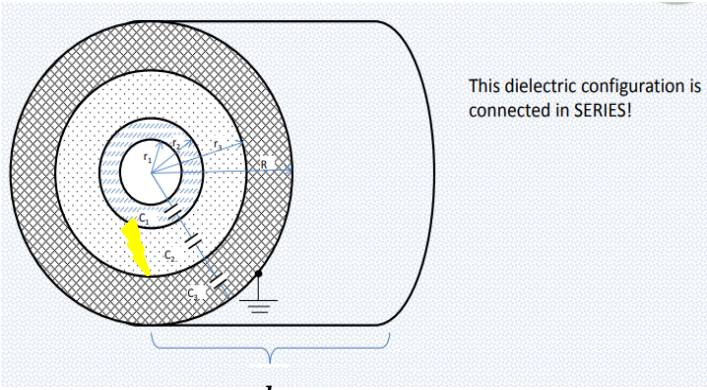
$$C\,U = C_1U_1 = C_2U_2 = C_3U_3 = \cdots = C_nU_n$$

$$U_1 = \frac{C\,U}{C_1}$$

$$U_2 = \frac{C\,U}{C_2}$$

$$\vdots$$

$$U_n = \frac{C\,U}{C_n}$$



3 MULTILAYER DIELECTRIC COAXIAL CYLINDERS

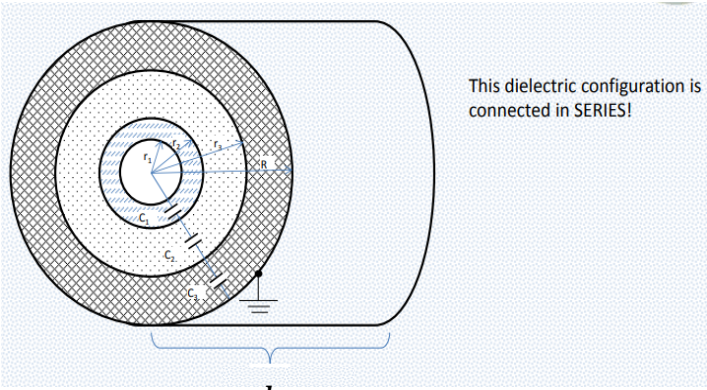
Equivalent capacitance of the system:

$$X_C = X_{C1} + X_{C2} + X_{C3} + \cdots + X_{Cn}$$

$$\frac{1}{\omega C} = \frac{1}{\omega C_1} + \frac{1}{\omega C_2} + \frac{1}{\omega C_3} + \cdots + \frac{1}{\omega C_n}$$

$$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \cdots + \frac{1}{C_n}$$

C : equivalent capacitance of the system, total capacitance of the system
U : voltage applied to the entire system
C_i : capacitance of the ith dielectric layer
U_i : voltage across i^{th} dielectric layer



3 MULTILAYER DIELECTRIC COAXIAL CYLINDERS

Calculating capacitance:

$$C_1 = 2\pi\varepsilon_1 l \frac{1}{\ln\frac{r_2}{r_1}}, C_2 = 2\pi\varepsilon_2 l \frac{1}{\ln\frac{r_3}{r_2}}, \dots, C_n = 2\pi\varepsilon_n l \frac{1}{\ln\frac{R}{r_n}}$$

$$\frac{1}{C} = \frac{1}{2\pi\varepsilon_1 l \frac{1}{\ln\frac{r_2}{r_1}}} + \frac{1}{2\pi\varepsilon_2 l \frac{1}{\ln\frac{r_3}{r_2}}} + \cdots + \frac{1}{2\pi\varepsilon_n l \frac{1}{\ln\frac{R}{r_n}}}$$

$$= \frac{1}{2\pi l}\left(\frac{1}{\varepsilon_1}\ln\frac{r_2}{r_1} + \frac{1}{\varepsilon_2}\ln\frac{r_3}{r_2} + \cdots + \frac{1}{\varepsilon_n}\ln\frac{R}{r_n}\right) = \frac{1}{2\pi l}\underbrace{\sum_{i=1}^{n}\frac{1}{\varepsilon_i}\ln\frac{r_{i+1}}{r_i}}_{A}$$

3 MULTILAYER DIELECTRIC COAXIAL CYLINDERS

Total capacitance, equivalent capacitance:

$$A = \sum_{i=1}^{n} \frac{1}{\varepsilon_i} \ln \frac{r_{i+1}}{r_i} = \frac{1}{\varepsilon_1} \ln \frac{r_2}{r_1} + \frac{1}{\varepsilon_2} \ln \frac{r_3}{r_2} + \cdots + \frac{1}{\varepsilon_n} \ln \frac{R}{r_n}$$

$$\frac{1}{C} = \frac{A}{2\pi l} \Rightarrow C = \frac{2\pi l}{A}$$

3 MULTILAYER DIELECTRIC COAXIAL CYLINDERS

Potantial and electric field across 1st dielectric layer:

$$U_1 = \frac{C\,U}{C_1} = \frac{\frac{2\pi l}{A}U}{2\pi\varepsilon_{\,1} l\,\frac{1}{\ln\frac{r_2}{r_1}}} = \frac{U}{A\varepsilon_{\,1}}\ln\frac{r_2}{r_1}$$

$$E_{1max} = \frac{U_1}{r_1 \ln\frac{r_2}{r_1}} = \frac{\frac{U}{A\varepsilon_{\,1}}\ln\frac{r_2}{r_1}}{r_1 \ln\frac{r_2}{r_1}} = \frac{U}{A\varepsilon_1 r_1}$$

$$E_{1min} = \frac{U_1}{r_2 \ln\frac{r_2}{r_1}} = \frac{\frac{U}{A\varepsilon_{\,1}}\ln\frac{r_2}{r_1}}{r_2 \ln\frac{r_2}{r_1}} = \frac{U}{A\varepsilon_1 r_2}$$

We can write the same equations with respect to applied voltage (U)

3 MULTILAYER DIELECTRIC COAXIAL CYLINDERS

Potantial and electric field across i th dielectric layer:

$$U_i = \frac{C.U}{C_i} = \frac{U}{A.\varepsilon_i.l_i}.\ln\frac{r_{i+1}}{r_i}$$

Voltage across i^{th} layer

Applied Voltage

$$E_{imax} = \frac{U_i}{r_i.\ln\frac{r_{i+1}}{r_i}} = \frac{U}{A.\varepsilon_i.r_i}$$

$$E_{imin} = \frac{U_i}{r_{i+1}.\ln\frac{r_{i+1}}{r_i}} = \frac{U}{A.\varepsilon_i.r_{i+1}}$$

3 MULTILAYER DIELECTRIC COAXIAL CYLINDERS

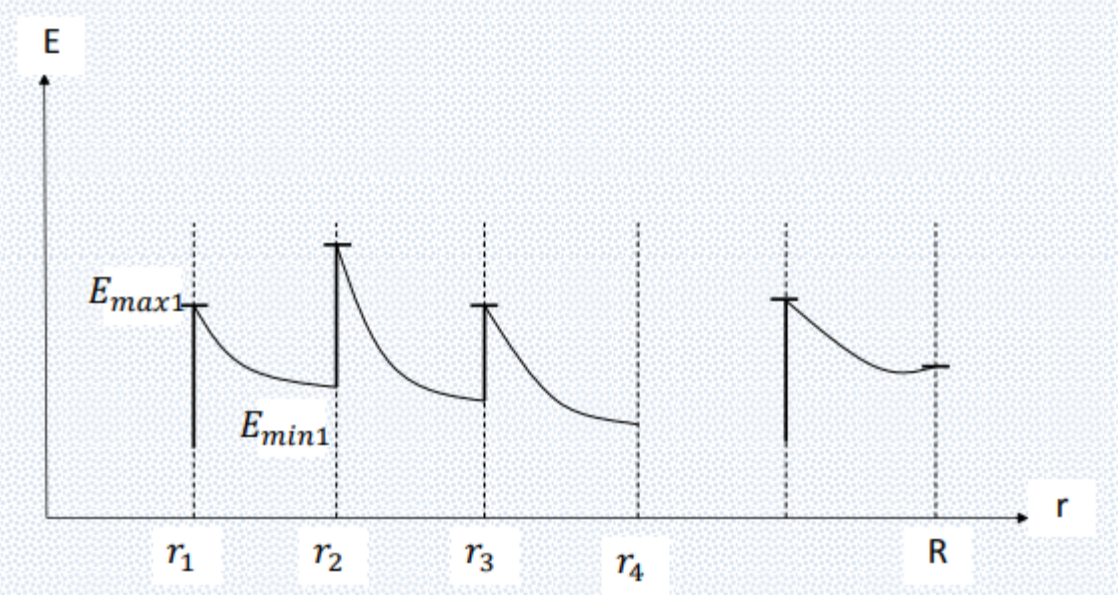
If the maximum electric field strength across any dielectric layer, $E_{i,max}$ becomes greater than breakdown strength of that dielectric material, there will be an electrical discharge.

$E_{i,max} \geq E_{bi} \rightarrow$ Discharge in that layer of the insulation!!!

Breakdown, short circuit in i^{th} dielectric layer

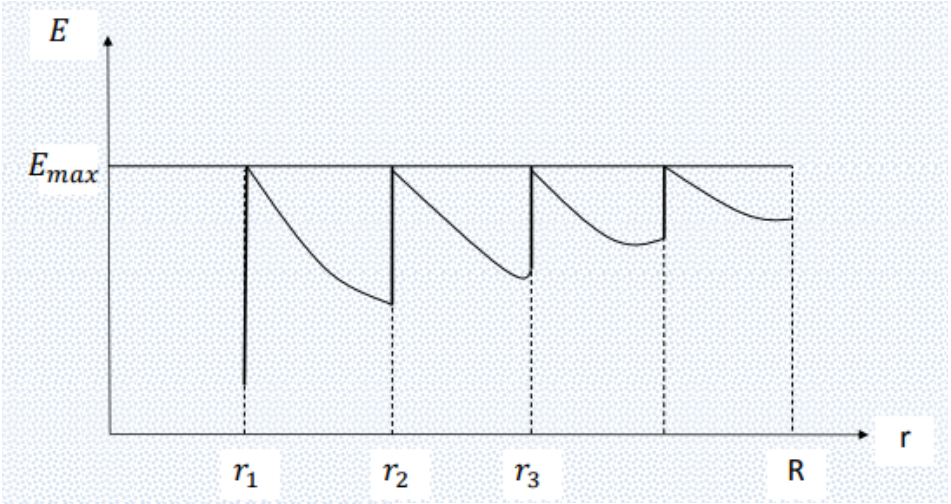
3 MULTILAYER DIELECTRIC COAXIAL CYLINDERS

Electric field distribution:



3 MULTILAYER DIELECTRIC COAXIAL CYLINDERS

Uniform stress condition:



To make the field disribution more uniform; (uniformly stressed)

$$E_{1max} = E_{2max} = \cdots = E_{nmax}$$

$$\frac{U}{A.\varepsilon_1.l_1.r_1} = \frac{U}{A.\varepsilon_2.l_2.r_2} = \cdots = \frac{U}{A.\varepsilon_n.l_n.r_n} \rightarrow \varepsilon_1.l_1.r_1 = \varepsilon_2.l_2.r_2 = \cdots = \varepsilon_n.l_n.r_n$$

If $l_1 = l_2 = \cdots = l_n = l = constant$

$$\varepsilon_1.r_1 = \varepsilon_2.r_2 = \cdots = \varepsilon_n.r_n$$

BREAKDOWN BEHAVIOR IN CYLINDRICAL ELECTRODE SYSTEMS

Actual electrode seperation, $d = r_2 - r_1$

$$E_{ave} = \frac{U}{d} \rightarrow d = \frac{U}{E_{ave}}$$

Effective electrode seperation, $\alpha = \frac{U}{E_{max}} = r_1 \ln \frac{r_2}{r_1}$

$$E_{max} = \frac{U}{\alpha} \rightarrow U = E_{max}.\alpha$$

Geometric characteristics, $p = \frac{r_1 + d}{r_1} = \frac{r_2}{r_1} = q$

Utilization factor (electrode efficiency), $\eta = \frac{E_{ave}}{E_{max}} = \frac{U/(r_2 - r_1)}{U/r_1 \ln \frac{r_2}{r_1}} = \frac{\alpha}{d} = \frac{r_1.lnp}{r_2 - r_1} = \frac{1}{p-1} lnp$

4 BREAKDOWN BEHAVIOR IN CYLINDRICAL ELECTRODE SYSTEMS

Analyzing the relationship between electrode radius, electric field and potential:

- Change in E_{max} with respect to r_1
- Change in U_{max} with respect to r_1

BREAKDOWN BEHAVIOR IN CYLINDRICAL ELECTRODE SYSTEMS

Change in E_{max} with respect to r_1

Keeping r_2 and U constant

Radius of outer cylinder is constant, radius of inner cylinder is variable

$E_{max} = \frac{U}{r_1 ln\frac{r_2}{r_1}}$ E_{max} depends on r_1

BREAKDOWN BEHAVIOR IN CYLINDRICAL ELECTRODE SYSTEMS

Change in E_{max} with respect to r_1

How can we find the minimum value for E_{max}? It is the case when its derivative is zero.

$$\frac{dE_{max}}{dr_1} = 0 \rightarrow$$

$$E_{max} = \frac{U}{r_1 . \ln\frac{r_2}{r_1}} = \frac{U}{r_1 . \ln r_2 - r_1 . \ln r_1}$$

$$\frac{dE_{max}}{dr_1} = \frac{-(\ln r_2 - \ln r_1 - 1).U}{(r_1 . \ln r_2 - r_1 . \ln r_1)^2} = 0$$

$$\ln r_2 - \ln r_1 - 1 = 0$$

$$\ln\frac{r_2}{r_1} = 1 = \ln e \rightarrow \frac{r_2}{r_1} = e$$

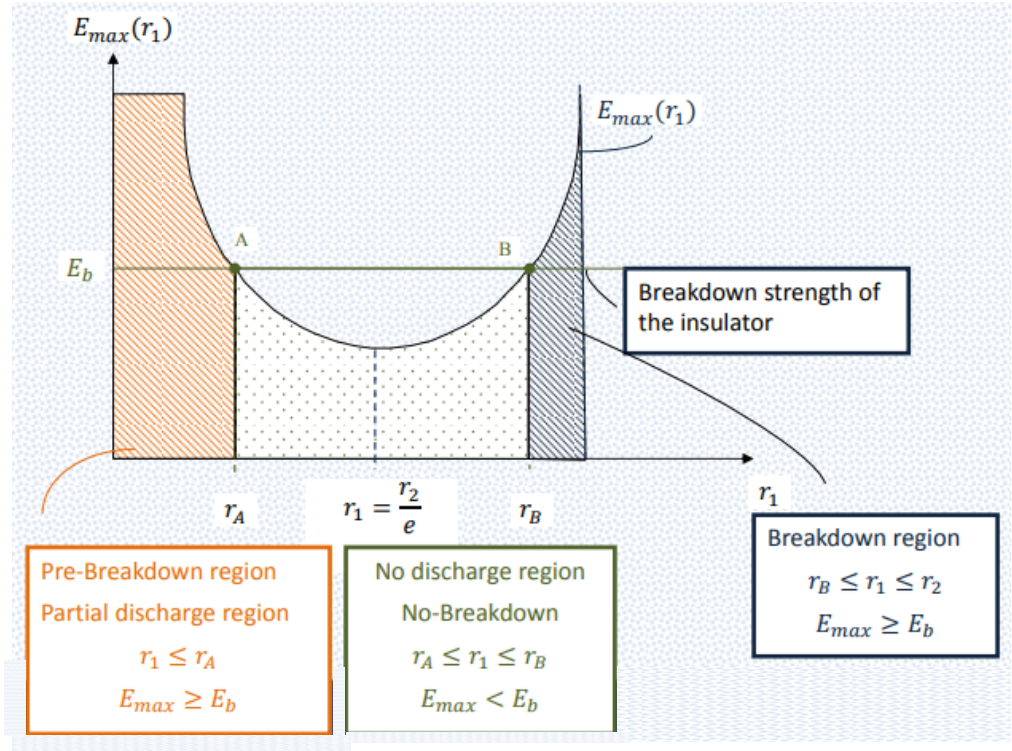
$$\frac{r_2}{r_1} = p = e = 2.718$$

$p = p_b$

fittest, optimum geometric char. with regard to breakdown in co-axial cylinders

4 BREAKDOWN BEHAVIOR IN CYLINDRICAL ELECTRODE SYSTEMS

Change in E_{max} with respect to r_1



BREAKDOWN BEHAVIOR IN CYLINDRICAL ELECTRODE SYSTEMS

Change in U_{max} with respect to r_1

Keeping r_2 and E_b constant

$$E_{max} = \frac{U}{r_1 ln \frac{r_2}{r_1}} = \frac{U}{\alpha}$$

$$U = \alpha . E_{max}$$

$\alpha = \alpha_{max}$ $\quad$ $U = U_{max}$ $\quad$ $E_{max} = E_b$

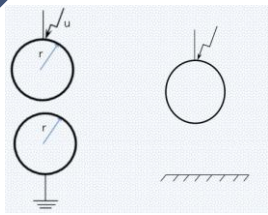
$$U_{max} = \alpha_{max} . E_b$$

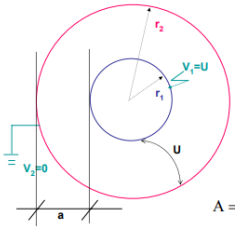
$$\alpha = r_1 . \ln \frac{r_2}{r_1} = f(r_1)$$

Maximum ⟶ Voltage will be **Maximum**

EE4047 – High Voltage Techniques I

Continuation to:
Cylindrical Electrode Systems





Week 10.2 – Monday Lesson

Asst. Prof. Dr. M. Alparslan Zehir

11.05.2023

OUTLINE

2 Multilayer Dielectric Concentric Spheres

OUTLINE

2 Multilayer Dielectric Concentric Spheres

BREAKDOWN BEHAVIOR IN CYLINDRICAL ELECTRODE SYSTEMS

Change in U_{max} with respect to r_1

Keeping r_2 and E_b constant

$$E_{max} = \frac{U}{r_1 ln \frac{r_2}{r_1}} = \frac{U}{\alpha}$$

$$U = \alpha . E_{max}$$

$\alpha = \alpha_{max}$ $\quad$ $U = U_{max}$ $\quad$ $E_{max} = E_b$

$$U_{max} = \alpha_{max} . E_b$$

$$\alpha = r_1 . \ln \frac{r_2}{r_1} = f(r_1)$$

Maximum ⟶ Voltage will be **Maximum**

BREAKDOWN BEHAVIOR IN CYLINDRICAL ELECTRODE SYSTEMS

Change in U_{max} with respect to r_1

How can we find the maximum value for α? It is the case when its derivative is zero.

$$\frac{d\alpha}{dr_1} = 0;\ \alpha \rightarrow \alpha_{max}$$

$$\alpha = r_1.\ln\frac{r_2}{r_1} = r_1.\ln r_2 - r_1.\ln r_1$$

$$\frac{d\alpha}{dr_1} = \ln r_2 + 0 - \ln r_1 - \frac{r_1}{r_1} = 0$$

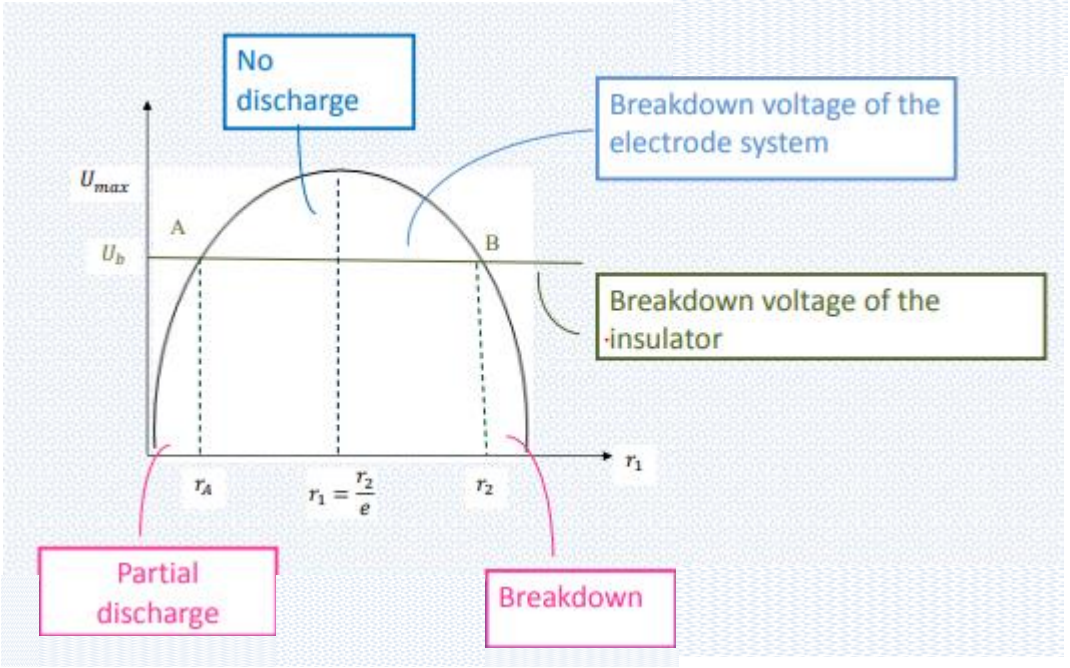
$$\rightarrow \ln r_2 - \ln r_1 - 1 = 0 \rightarrow \ln\frac{r_2}{r_1} = 1$$

$$\frac{r_2}{r_1} = e = 2.718$$

Optimum characteristic with regard to breakdown in cylindrical system.

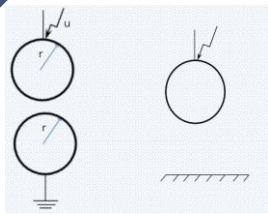
BREAKDOWN BEHAVIOR IN CYLINDRICAL ELECTRODE SYSTEMS

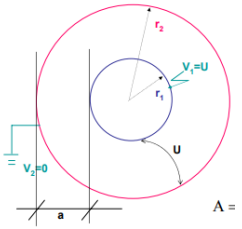
Change in U_{max} with respect to r_1



EE4047 – High Voltage Techniques I

New Topic:
Electrical Breakdown in Gases





Week 10.2 – Thursday Lesson

Asst. Prof. Dr. M. Alparslan Zehir

11.05.2023

OUTLINE

2 Multilayer Dielectric Concentric Spheres

OUTLINE

1 Some Basic Concepts

2 Introduction to Breakdown in Gases

3 Ionization Types in Gases

4 Discharge theorems

1 SOME BASIC CONCEPTS

Insulation

To prevent electrical conduction between points with different potentials

Insulator

Material used for obtaining electrical insulation in the insulation system

- Gas
- Liquid
- Solid
- Vacuum

1 SOME BASIC CONCEPTS

Electrical discharge

Any dielectric (or insulator) can withstand a voltage applied below a critical value.

If applied voltage across dielectric exceeds this critical value, a discharge occurs in the insulator medium.

1 SOME BASIC CONCEPTS

Electrical discharge

There are different types of electrical discharge.

- **Breakdown:** Complete electrical discharge through the insulator
- **Flashover:** Complete electrical discharge jumping around the insulator
- **Partial discharge:** Partial breakdown on conductors at points with the highest electrical stress

2 INTRODUCTION TO BREAKDOWN IN GASES

Gases as insulating medium is often preferred in high voltage technique application because of their self-restoring capability after a breakdown.

The most common gases are air and SF_6.

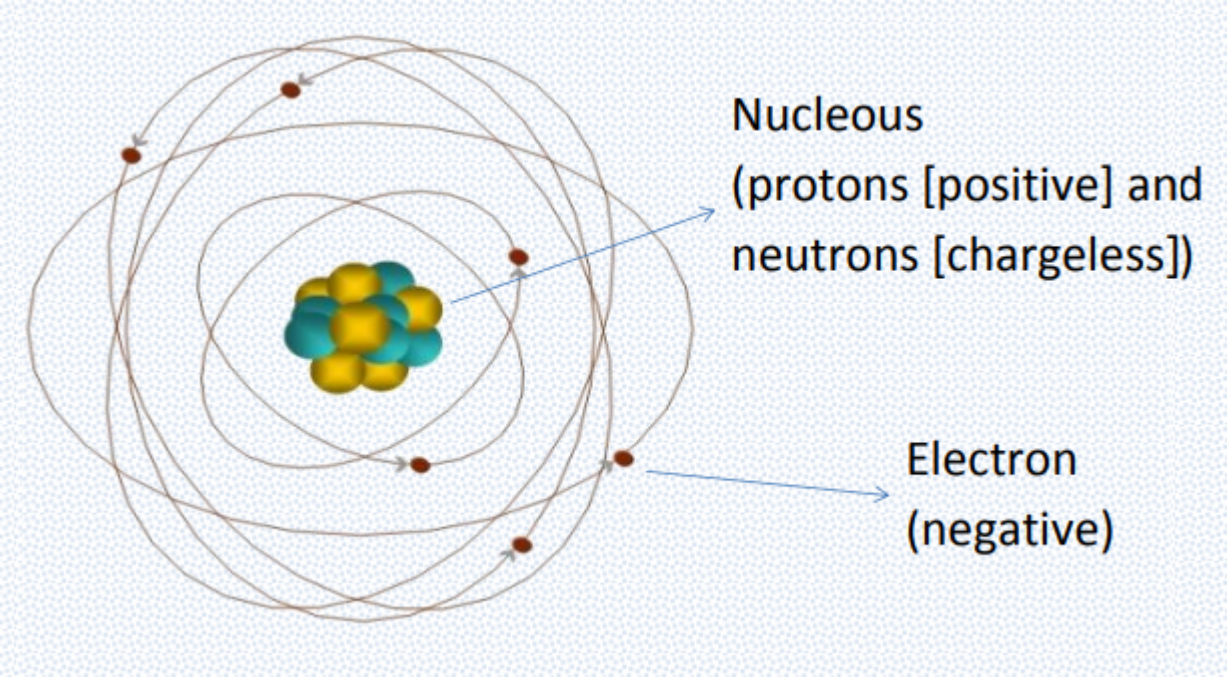
Discharge in gases

Generally, a neutral gas does not conduct electricty when it is conserved from external factors. However, application of electric field can lead the gas to lose its insulating properties at a critical value.

This conduction in dielectric is called ELECTRICAL DISCHARGE. Electrical discharge occurs in the presence of charged particles which are electrons (negatively charged particles) and ions (negatively or positively charged atoms or molecules).

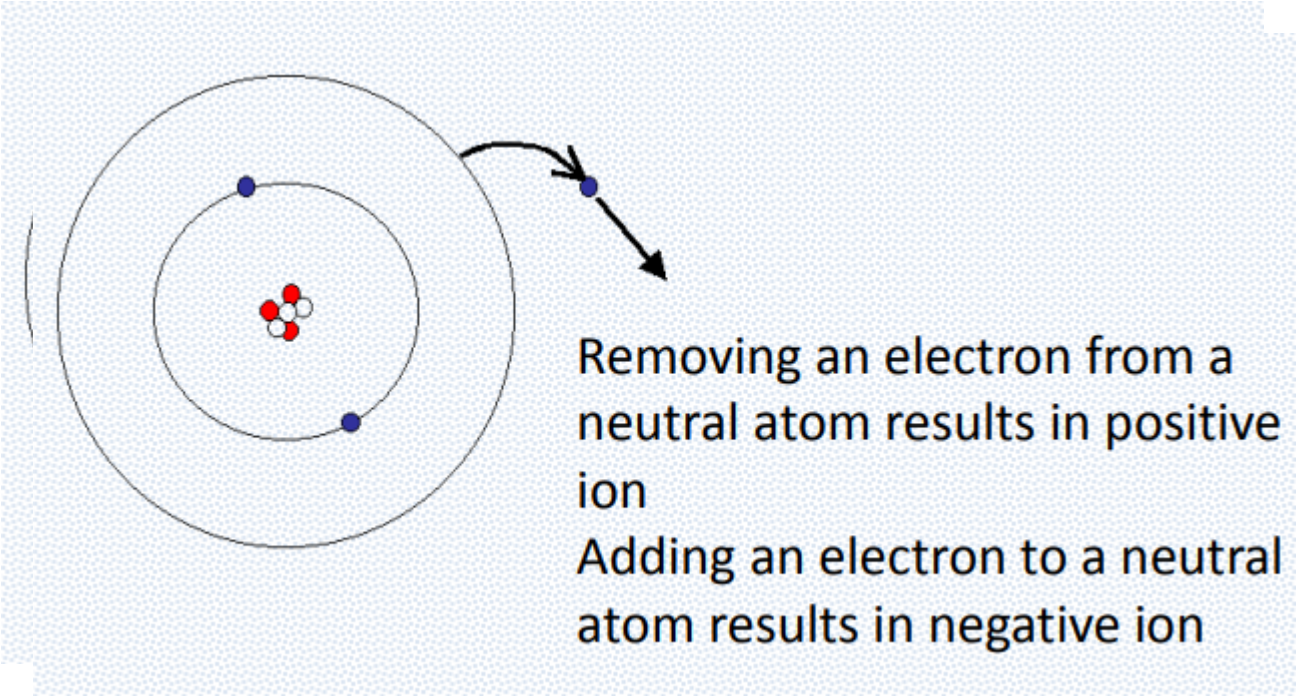
INTRODUCTION TO BREAKDOWN IN GASES

Structure and excitation of atom and ionization



INTRODUCTION TO BREAKDOWN IN GASES

Structure and excitation of atom and ionization



INTRODUCTION TO BREAKDOWN IN GASES

Energy on a charge

$$F = q\,E$$

$$E = \frac{U}{d}$$

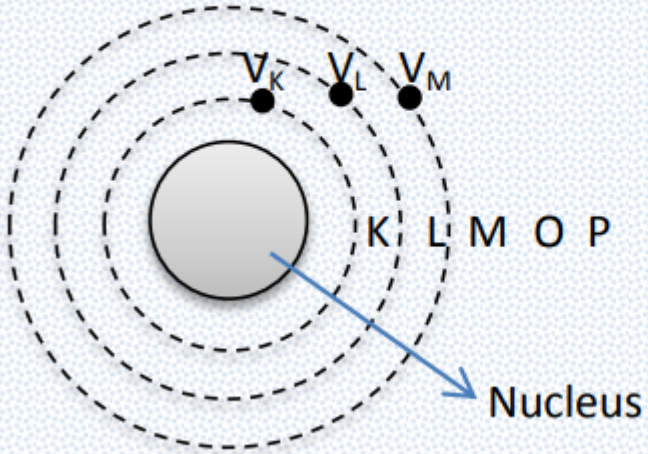
$$F = q\frac{U}{d}$$

$$Fd = qU = W\ (energy, work)$$

INTRODUCTION TO BREAKDOWN IN GASES

Excitation energy

$$W_e = q\,U_e$$



$$W_e = q\,(V_K - V_L)$$

When a voltage between two orbits is applied to an electron, electron jumps to the one orbit level up.

Photon energy

The electron stays at this new orbit level for about $10^{-8} - 10^{-9}$ seconds and it return to its original position.

The electron emits energy as photon (beam) energy.

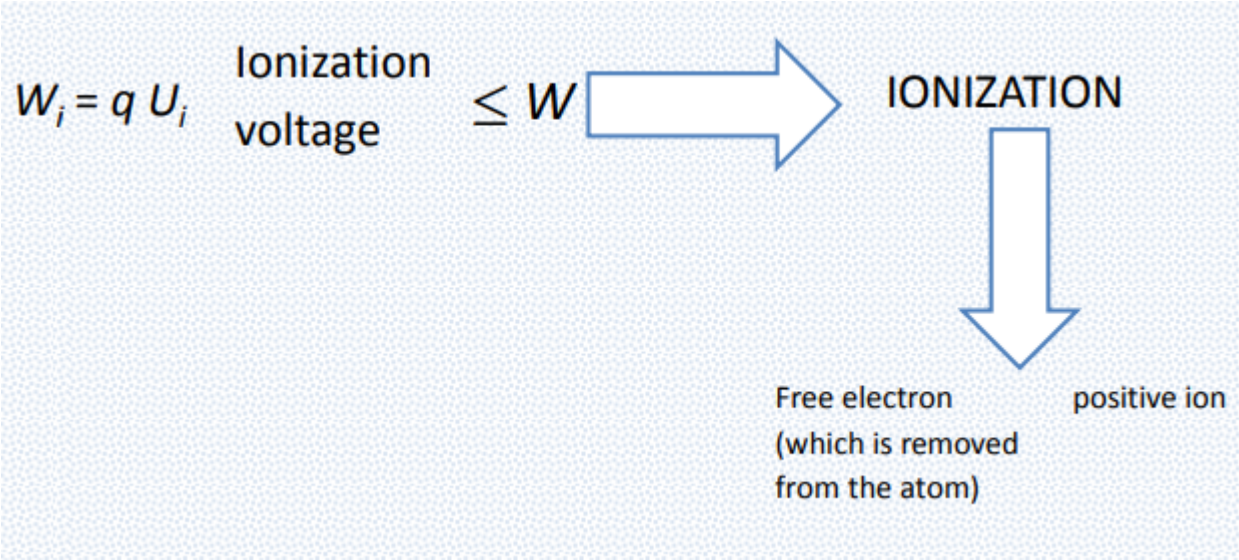
$W_e = qU_e = h\,f_e$ Photon energy (quantum of energy)

h: Planck's constant; f_e: frequency of the radiation

3 IONIZATION TYPES IN GASES

Ionization

Removing an electron from an atom or a molecule is called ionization.



3 IONIZATION TYPES IN GASES

1. Ionization by collision

Gas atoms and molecules collect kinetic energy by moving. When atoms or molecules collide, they transfer their energies to each other.

$$E_k = \frac{1}{2}mv^2 : \text{kinetic energy}$$

$$W_i \leq \frac{1}{2}mv^2 \rightarrow ionisation\ by\ collision$$

When kinetic energy of an electron exceeds the ionization energy of the atom or molecule (the energy needed to be given to an atom or a molecule in order to remove an electron), ionization occurs. Sometimes ionization occurs progressively.

3 IONIZATION TYPES IN GASES

2. Photo-ionization

A molecule in the ground state can be ionized by a photon having frequency f provided that the quantum energy emitted h (by an electron jumping from one orbit to another), is greater than the ionization energy of the molecule (gas gains photon's energy).

$$W_p = h\, f_p : \text{Photon energy}$$

h: Planck's constant = 6.625×10^{-34} Joule.s; f_p: frequency of photon

3 IONIZATION TYPES IN GASES

2. Photo-ionization

$f = \frac{v}{\lambda}$, $\lambda \downarrow \rightarrow f \uparrow \rightarrow W = h f \uparrow$ $\lambda = \frac{c}{f}$

(v: speed & λ: wavelength)

$$W = h f \geq W_i \rightarrow photoionisation$$

Photon with a small wavelength (λ) or high frequency (f) can supply enough energy for ionization.

3 IONIZATION TYPES IN GASES

3. Thermal ionization

3.1. Gas molecules move faster when heated.

$$E_k = \frac{1}{2}mv^2 : \text{kinetic energy increases}$$

$$W_i \leq \frac{1}{2}mv^2 \rightarrow ionisation\ by\ collision$$

3.2. Gas emits light when heated

$$W = h\,f \geq W_i \rightarrow photoionisation$$

3 IONIZATION TYPES IN GASES

4. Surface ionization

4.1. Electron Bombardment:

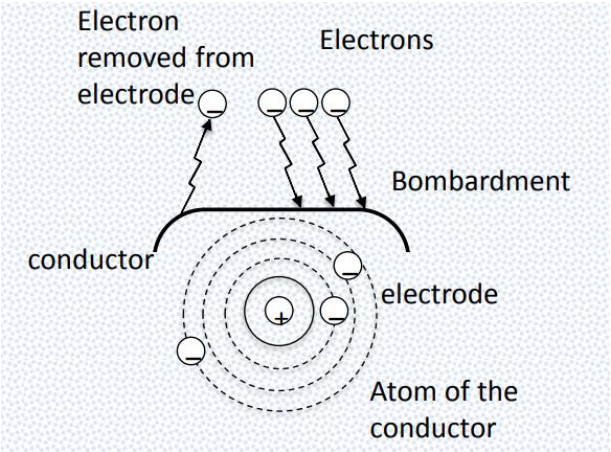
When electrons or positive ions impinge to the electrode, they will remove electron.

Minimum energy that has to be supplied to the conductor atom or molecule is called <u>output energy</u>

$$W_o \ll W_i = 2 - 25\ eV$$

W_o: 0.1 – 1 eV, Output energy of the electrode

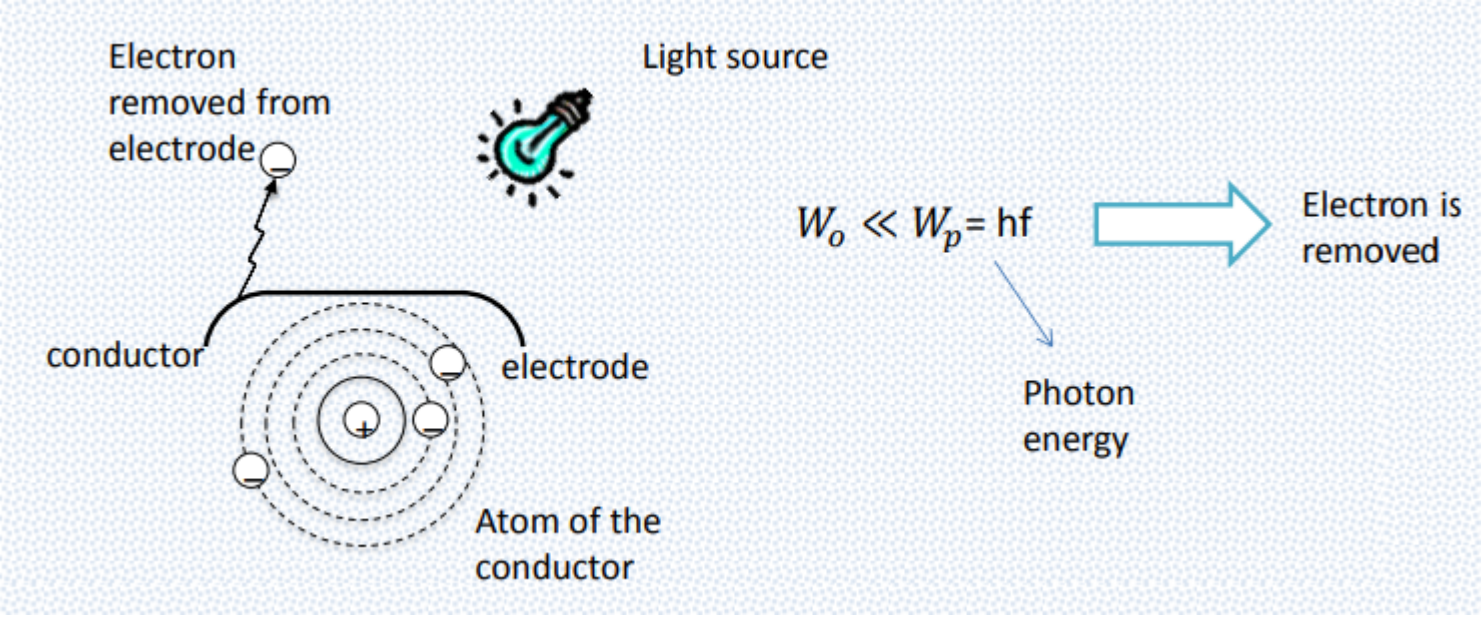
W_i: ionization energy of the gas



3 IONIZATION TYPES IN GASES

4. Surface ionization

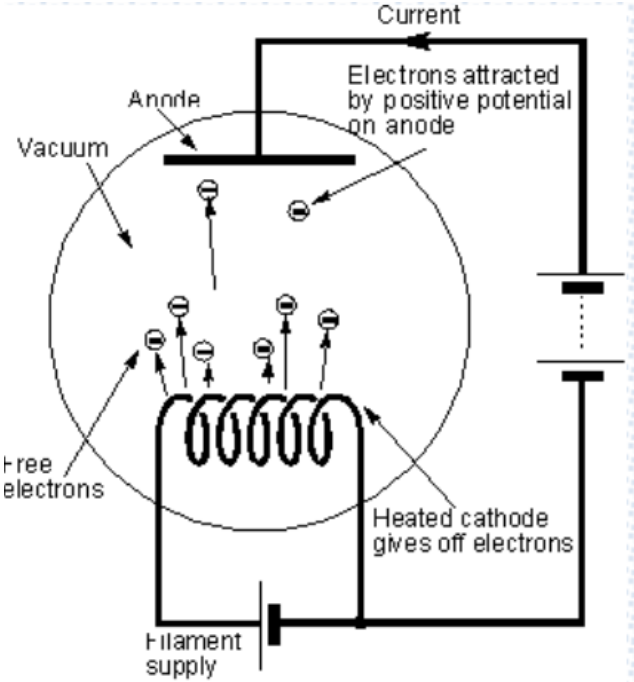
4.2. Surface ionization by light:



3 IONIZATION TYPES IN GASES

4. Surface ionization

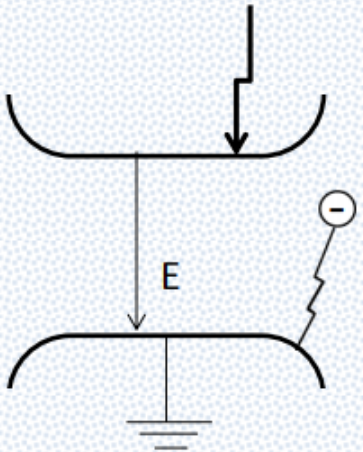
4.3. Surface ionization by heat:



IONIZATION TYPES IN GASES

4. Surface ionization

4.4. Surface ionization by field effect:



F = q E

E = U/d

F = q U/d

$W_{o,atom} = 1 - 5\ eV$

$W_{i,gas} = 2 - 25\ eV$

$W_o < W_{i,gas}$

it is easier to remove an electron from metal rather than gas!!!

1 DISCHARGE THEOREMS

1. Townsend Theorem

Electron Avalanche mechanism:

Suppose a free electron exists (due to cosmic radiation or some other external effect) in a gas where an electric field exists.

If E is sufficiently high, ionization by collision is very possible.

If we apply E, free electrons are accelerated
- Kinetic energy
- in eV
- move towards the anode

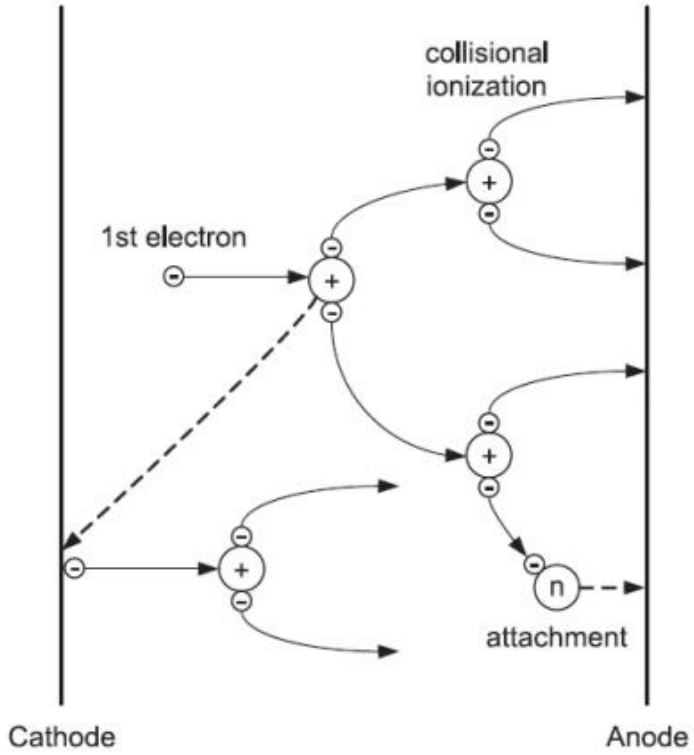
1 DISCHARGE THEOREMS

1. Townsend Theorem

Collision:

This process is cumulative.

The number of electrons will go on increasing as they continue to move under the action of E.



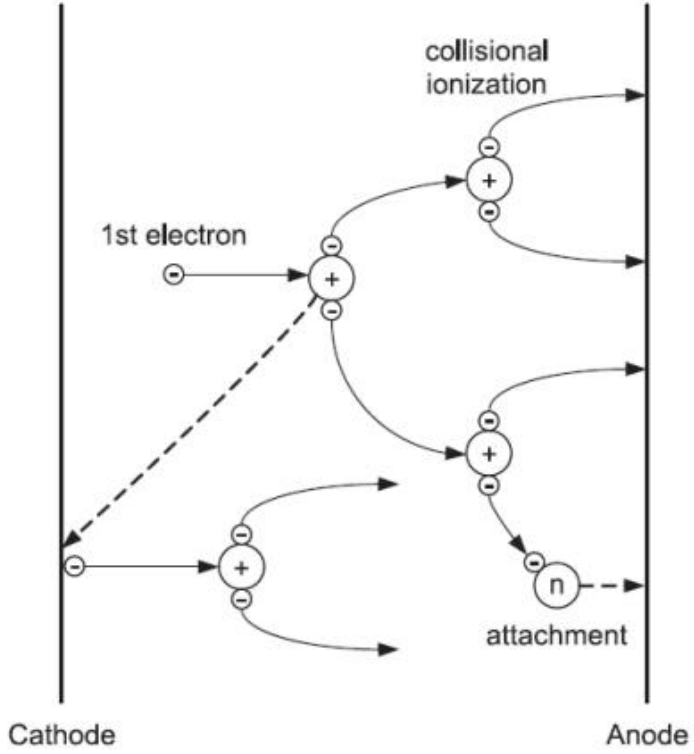
1 DISCHARGE THEOREMS

1. Townsend Theorem

Collision:

This process is cumulative.

The number of electrons will go on increasing as they continue to move under the action of E.



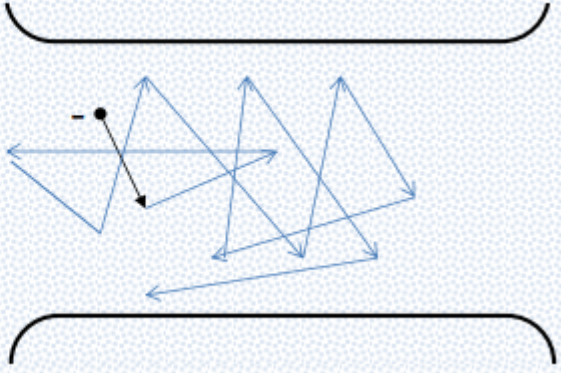
1 DISCHARGE THEOREMS

1. Townsend Theorem

Free Path law:

Free Path (λ) is the distance traveled by a particle between two collisions.

Random movements of gas atoms or molecules



1 DISCHARGE THEOREMS

1. Townsend Theorem

Mean Free Path ($\lambda_{ave} = \bar{\lambda}$)

It is the arithmetic average distance travelled by a moving particle between successive impacts.
or
Arithmetic average distance travelled by more than one particle in a certain period of time.

$$\bar{\lambda} = \lambda_{ave} = \frac{1}{n}(\lambda_1 + \lambda_2 + \cdots + \lambda_n)$$

n: number of collision, or number of free paths

DISCHARGE THEOREMS

1. Townsend Theorem

Free Path law:

Probability of a particle to travel a distance equal or greater than λ

$$f(\lambda) = P(\lambda) = e^{-\frac{\lambda}{\lambda_{ave}}}$$

Probability of $\lambda \geq 0$ $\quad e^{-\frac{0}{\lambda_{ave}}} = e^{0} = 1 \rightarrow 100\%$

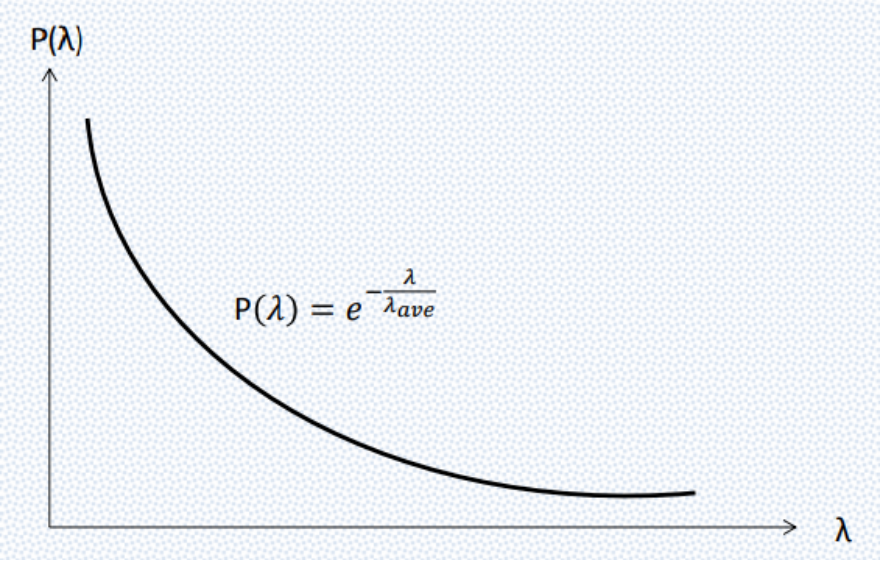
Probability of $\lambda \geq \lambda_{ave}$ $\quad e^{-\frac{\lambda_{ave}}{\lambda_{ave}}} = e^{-1} = \frac{1}{e} \rightarrow 37\%$

Probability of $\lambda \geq 3\lambda_{ave}$ $\quad e^{-\frac{3\lambda_{ave}}{\lambda_{ave}}} = e^{-3} = \frac{1}{e^{3}} \rightarrow 5\%$

1 DISCHARGE THEOREMS

1. Townsend Theorem

Probability function:



DISCHARGE THEOREMS

1. Townsend Theorem

- Number of initial particles: n_0
- Number of non-colliding particles: n_λ

$$n_\lambda = n_0 \, e^{-\frac{\lambda}{\lambda_{ave}}}$$

$$\begin{matrix} \textit{Number of} \\ \textit{total particles} \end{matrix} = \begin{matrix} \textit{number of} \\ \textit{colliding particles} \end{matrix} + \begin{matrix} \textit{number of} \\ \textit{noncolliding particles} \end{matrix}$$

1 DISCHARGE THEOREMS

1. Townsend Theorem

Free paths depending on particle properties:

$$\lambda_{ave} = \frac{1}{\pi\,(r_1 + r_2)^2\,N} \cdot \frac{v_{ave}}{\sqrt{{v_{1ave}}^2 + {v_{2ave}}^2}}$$

$r1$: radius ofthe 1st particle

$r2$: radius ofthe 2nd particle

v_{1ave}: average speed of the 1st particle

v_{2ave}: average speed of the 2nd particle

N: number of particles in unit volume (1 cm^3)

DISCHARGE THEOREMS

1. Townsend Theorem

Free paths depending on particle properties:
Molecule-molecule collision – mean free path of molecules:

$$r_1 = r_2 = r$$

$$v_{1ave} = v_{2ave} = v_{ave}$$

$$\lambda_{m,ave} = \frac{1}{\pi\,(2r)^2\,N} \cdot \frac{v_{ave}}{\sqrt{{v_{ave}}^2 + {va_{ve}}^2}}$$

$$= \frac{1}{4\sqrt{2}\pi r^2 . N}$$

DISCHARGE THEOREMS

1. Townsend Theorem

Free paths depending on particle properties:
Molecule-ion collision – mean free path of ion:

$$r_{ion} \cong r_{molecule} \cong r$$

$$v_{ion} > v_{molecule} \rightarrow ignore\ v_{molecule}$$

$$\lambda_{i,ave} = \frac{1}{\pi\,(2r)^2\,N} \cdot \frac{v_{ion}}{\sqrt{{v_{ion}}^2}} = \frac{1}{4\pi r^2 . N}$$

DISCHARGE THEOREMS

1. Townsend Theorem

Free paths depending on particle properties:
Molecule-electron collision – mean free path of electrons:

$$r_{electron} \ll r_{molecule} = r$$

$$v_{electron} \gg v_{molecule} \rightarrow ignore\ v_{molecule}$$

$$\lambda_{e,ave} = \frac{1}{\pi\, r^2\, N} \cdot \frac{v_{electron}}{\sqrt{{v_{electron}}^2}} = \frac{1}{\pi r^2 . N}$$

DISCHARGE THEOREMS

1. Townsend Theorem

Free paths depending on particle properties:
Comparison of mean free path of electron, ion and molecule:

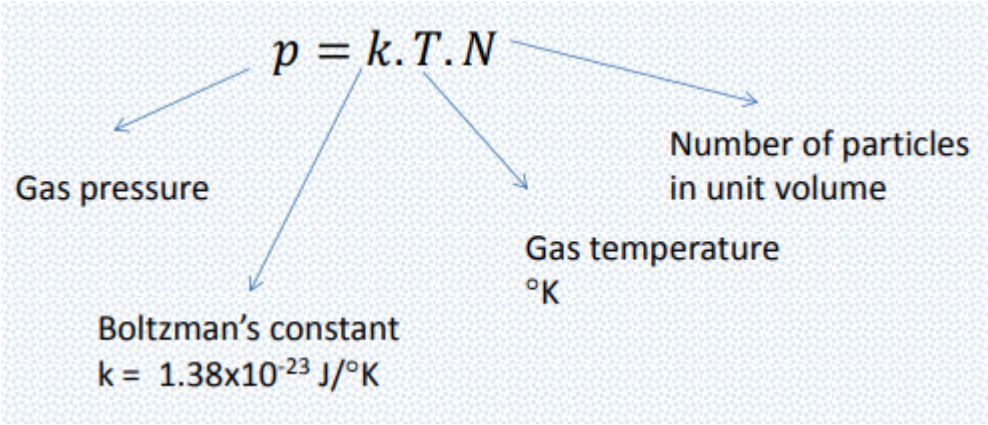
$$\lambda_{e,ave} = 4\lambda_{i,ave} = 4\sqrt{2}\lambda_{m,ave}$$

DISCHARGE THEOREMS

1. Townsend Theorem

Determination of Townsend's coefficient:

Due to gas general law:



DISCHARGE THEOREMS

1. Townsend Theorem

Determination of Townsend's coefficient:

For an electron:

$$\lambda_{e,ave} = \frac{1}{\pi r^2 . N}$$

Substitute this $N = \frac{p}{k.T}$

$$\lambda_{e,ave} = \frac{1}{\pi r^2 . \frac{p}{k.T}} = \frac{k.T}{\pi r^2 . p} = f(T, p)$$

$$\frac{k.T}{\pi r^2} = \frac{1}{A}$$

Constant!!! (if T is constant)

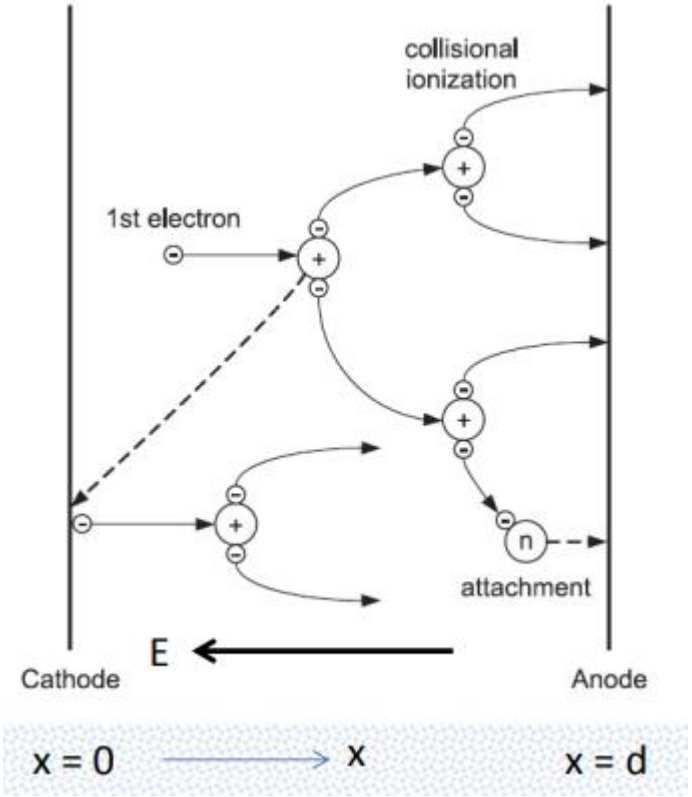
$$\lambda_{e,ave} = \frac{1}{A.p}$$

1 DISCHARGE THEOREMS

1. Townsend Theorem

Townsend's first ionization coefficient, α:

It is the number of ionizing collisions, on average, made by one electron per unit drift (1 cm) in the direction of the field.



DISCHARGE THEOREMS

1. Townsend Theorem

Townsend's first ionization coefficient, α:

$$\alpha = \underbrace{\frac{1}{\lambda_{e,ave}}}_{\text{Number of collisions}} \underbrace{e^{-\frac{\lambda_{e,i}}{\lambda_{e,ave}}}}_{\text{Probability of ionization in one collisison}}$$

$$\lambda_{e,ave} = \frac{1}{A.p}$$

$$\lambda_{e,i} = \frac{V_i}{E}$$

1 DISCHARGE THEOREMS

1. Townsend Theorem

Townsend's second ionization coefficient, γ

The first coefficient, α

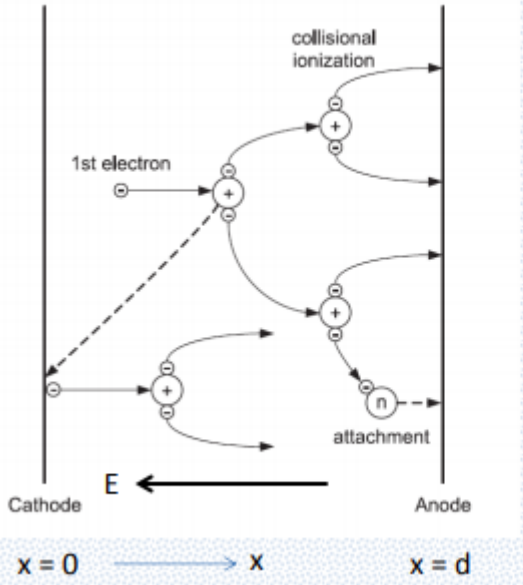
is the number of ion pairs made by one positive ion per unit drift (1 cm) in the direction of the field.

Positive ions removes electron from the cathode when they strike to cathode.

Townsend's ionization coefficient is about that phenomena, not about ionization by collision.

1 DISCHARGE THEOREMS

1. Townsend Theorem



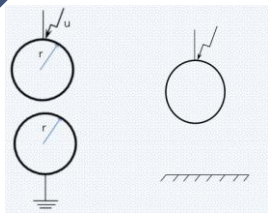
γ: is the number of electrons removed from the cathode by one positive ion striking to cathode.

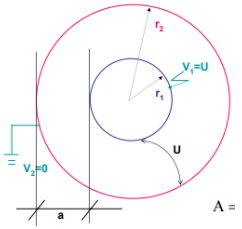
$$\gamma: \frac{1}{50}, \dots, \frac{1}{5000}$$

If γ =1/50, one electron can be removed from the cathode when 50 positive ions strike to it.

EE4047 – High Voltage Techniques I

Continuation to:
Electrical Breakdown in Gases





Week 11.2 – Thursday Lesson

Asst. Prof. Dr. M. Alparslan Zehir

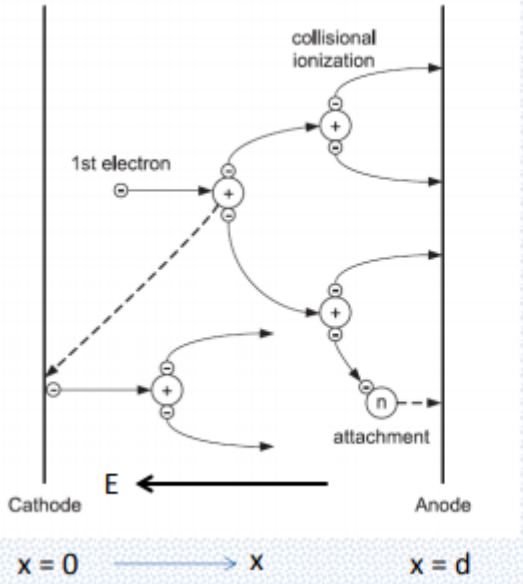
18.05.2023

OUTLINE

4 Continuation to: Discharge theorems

1 DISCHARGE THEOREMS

1. Townsend Theorem



γ: is the number of electrons removed from the cathode by one positive ion striking to cathode.

$$\gamma: \frac{1}{50}, \dots, \frac{1}{5000}$$

If γ =1/50, one electron can be removed from the cathode when 50 positive ions strike to it.

1 DISCHARGE THEOREMS

1. Townsend Theorem

$\gamma = 0.02 \sim 0.00002$

γ depends on the material of the cathode and the type os gaseous insulator.

Material of cathode	Gas	
	Air	Neon
Al	0.035	0.1
Cu	0.025	0.065
Fe	0.02	0.06

1 DISCHARGE THEOREMS

1. Townsend Theorem

At a distance x = d,

$(e^{\alpha d} - 1)$ positive ions are produced by collison

$\gamma(e^{\alpha d} - 1)$ electrons are produced

Number of electrons removed from the cathode by one positive ion

1 DISCHARGE THEOREMS

1. Townsend Theorem

$$\gamma\left(e^{\alpha d}-1\right)=1$$

If this condition is fulfilled, at least one electron is removed from the cathode and discharge can continue.

This is also called as; **condition of self-sustainability of discharge.**

1 DISCHARGE THEOREMS

1. Townsend Theorem

Equation for breakdown voltage in uniform fields by Townsend's theory (Paschen Equation)

$$U_b = \frac{B.p.d}{\ln\left[\frac{A.p.d}{\ln(1+\frac{1}{\gamma})}\right]}$$

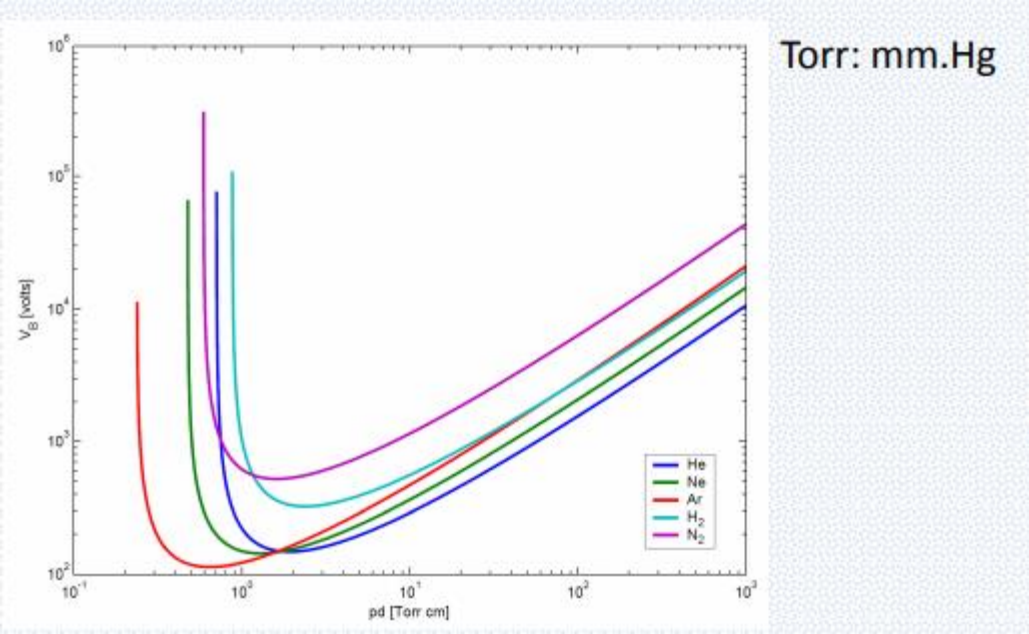
$$U_b = \mathrm{f(p.d)}$$

Breakdown voltage depends on multiplication of pressure and electrode gap

1 DISCHARGE THEOREMS

1. Townsend Theorem

Breakdown voltage curve (Paschen curve)



1 DISCHARGE THEOREMS

1. Townsend Theorem

The critical (p.d) value that makes the breakdown voltage minimum

$$(p.d)_{cr} = \frac{e.\ln(1+\frac{1}{\gamma})}{A}$$

1 DISCHARGE THEOREMS

1. Townsend Theorem

The critical (p.d) value that makes the breakdown voltage minimum

$$U_b[(p.d)_{cr}] = U_{b,min} = \frac{B.(p.d)_{cr}}{\ln\left[\frac{A.(p.d)_{cr}}{\ln(1+\frac{1}{\gamma})}\right]}$$

$$U_{b,min} = \frac{B.\frac{e.\ln(1+\frac{1}{\gamma})}{A}}{\ln\left[\frac{A.\frac{e.\ln(1+\frac{1}{\gamma})}{A}}{\ln(1+\frac{1}{\gamma})}\right]} = B.\underbrace{\frac{e.\ln(1+\frac{1}{\gamma})}{A}}_{(p.d)_{cr}}$$

$$U_{b,min} = B.(p.d)_{cr}$$

1 DISCHARGE THEOREMS

1. Townsend Theorem

The critical (p.d) value that makes the breakdown voltage minimum

$$E = {}^{U}/_{d}$$

$$E_b = {}^{U_b}/_{d}$$

$$E_b = \frac{B.p}{\ln\left[\frac{A.p.d}{\ln(1+\frac{1}{\gamma})}\right]}$$

1 DISCHARGE THEOREMS

1. Townsend Theorem

Generalized Pasched Equation

$A \rightarrow A.{}^{T}/_{T_0}$
$B \rightarrow B.{}^{T}/_{T_0}$

$$U_b = \frac{B.\frac{T}{T_0}.p.d}{\ln\left[\frac{A.\frac{T}{T_0}.p.d}{\ln(1+\frac{1}{\gamma})}\right]}$$

- $T_0 = 273 + 20°C = 293°K$
- $T = 273 + \theta°C$: gas temperature in $°K$
- p: gas pressure in mmHg
- A, B: gas constants
- γ: number of electrons removed from the cathode by positive ions

1 DISCHARGE THEOREMS

Streamer Theory

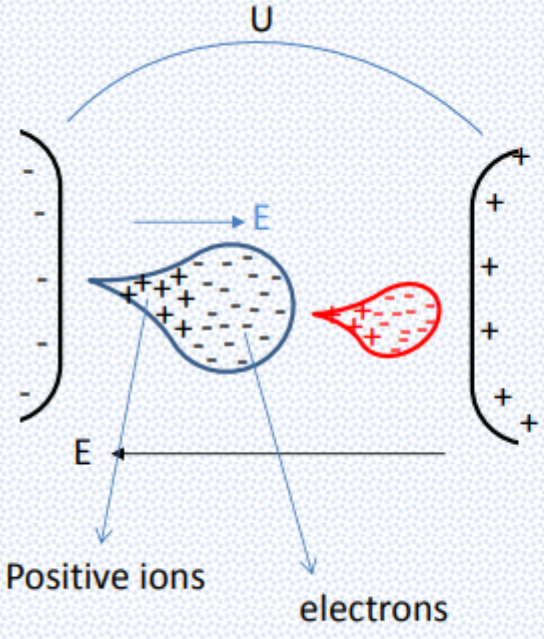
This theory explains electrical discharges in big gaps and high pressure.

p.d>500 mmHg.cm

The Streamer Theory predicts the development of a spark discharge directly from a single avalanche.

DISCHARGE THEOREMS

Streamer Theory



the electrons advance rapidly, the positive ions are left behind in a relatively slow-moving tail

Electric field between the avalanche and the anode is big

DISCHARGE THEOREMS

Streamer process

• The space charge produced in the avalanche causes sufficient distortion of the electric field that those free electrons move towards the avalanche head, and in so doing generate further avalanches in a process that rapidly becomes cumulative.
• As the electrons advance rapidly, the positive ions are left behind in a relatively slow-moving tail
• The field will be enhanced in front of the head. Just behind the head the field between the electrons and the positive ions is in the opposite direction to the applied field and hence the resultant field strength is less.
• Again between the tail and the cathode the field is enhanced.
• Due to the enhanced field between the head and the anode, the space charge increases, causing a further enhancement of the field around the anode.
• The process is very fast and the positive space charge extends to the cathode very rapidly resulting in the formation of a streamer.

DISCHARGE THEOREMS

Breakdown voltage by Streamer Theory

$$U_b = 5.27 \times 10^{-7} \frac{\alpha . d . e^{\alpha d}}{\sqrt{\frac{d}{p}}}$$

U_b: V

α: 1/cm

d: cm

p: mmHg

d: electrode gap [cm] (electrode seperation)

p: gas pressure [mmHg]

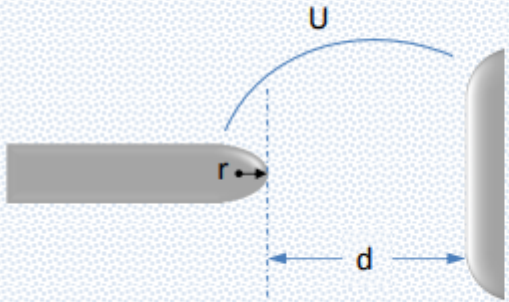
α: ionization coefficient of the eletron [1/cm]

U_b: breakdown voltage [V]

$$E_b = \frac{U_b}{d} = 5.27 \times 10^{-7} \frac{\alpha . e^{\alpha d}}{\sqrt{\frac{d}{p}}}$$

DISCHARGE THEOREMS

Streamer Theory for non-uniform electric field



r: curvature radius

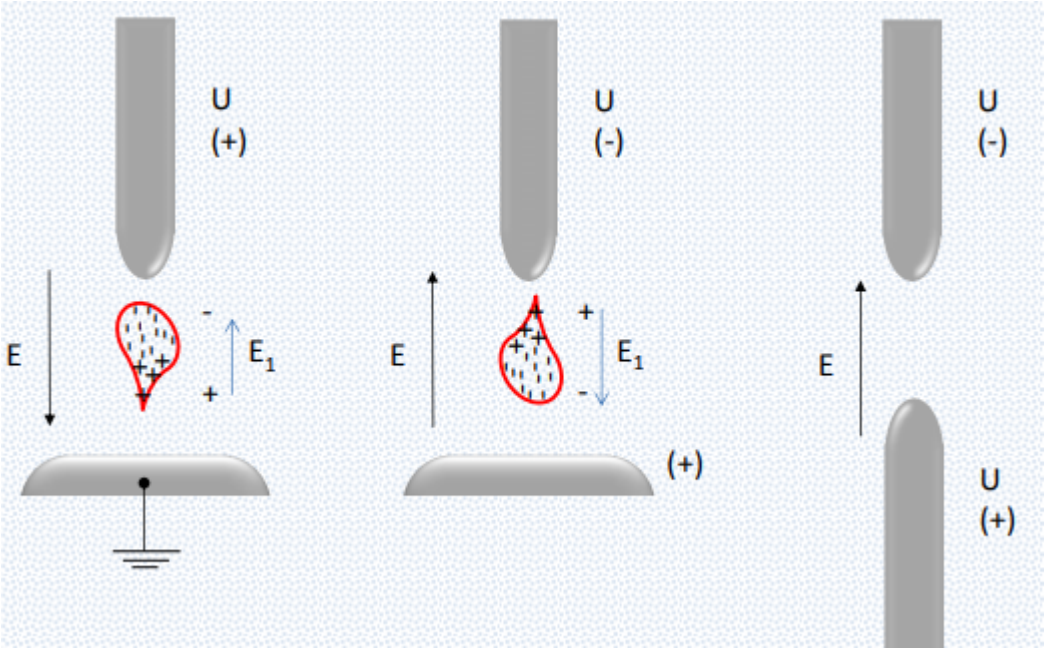
d: electrode gap

Non-uniform field:

$$\frac{d}{r} > 40 \rightarrow d > 40\,r$$

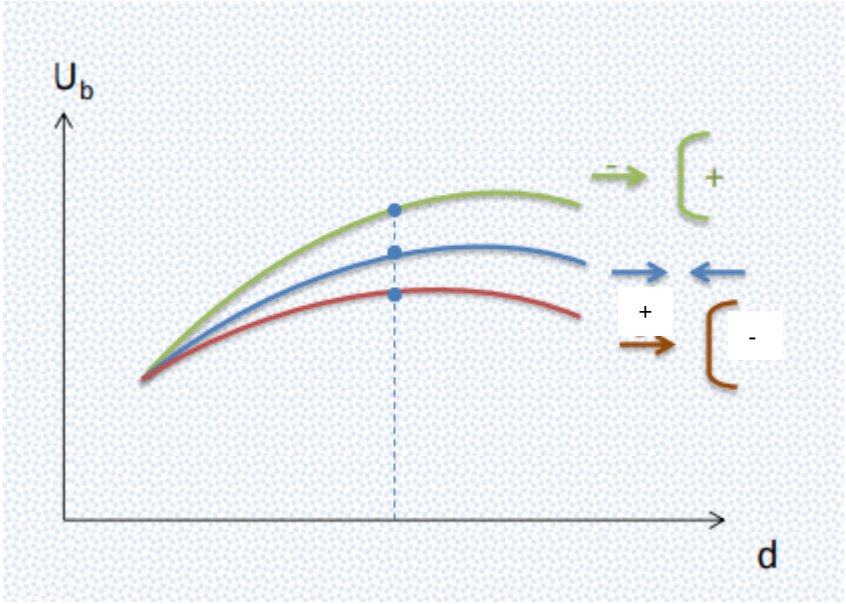
DISCHARGE THEOREMS

Different non-uniform electric field cases



DISCHARGE THEOREMS

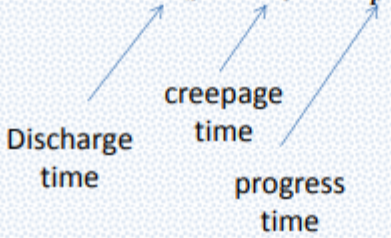
Electrode gap – breakdown voltage



1 DISCHARGE THEOREMS

Discharge time

$$t_b = t_\sigma + t_p$$



Creepage time: time required for an electron to appear after the moment which voltage applied

Progress time: time required for such an electron appeared in the gap to lead a breakdown after the moment it appeared

EE4047 – High Voltage Techniques I

New Topic:
Corona Discharges

Week 11.2 – Thursday Lesson

Asst. Prof. Dr. M. Alparslan Zehir

18.05.2023

OUTLINE

1 Definition

2 Characteristics Properties

3 Mechanism of Corona

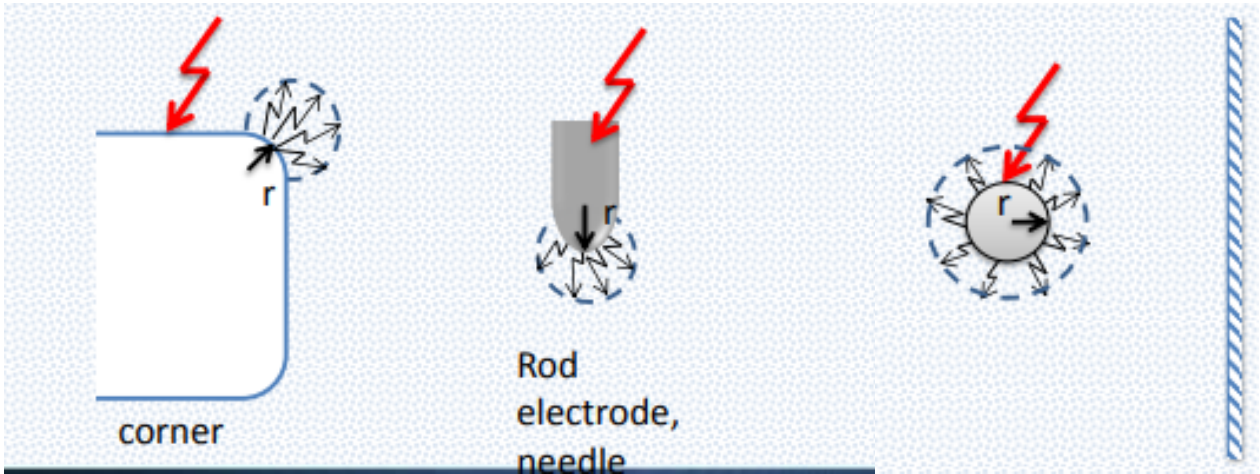
4 Power Loss due to Corona

1 DEFINITION

Corona

Literal meaning of corona is ‘’light crown’’.

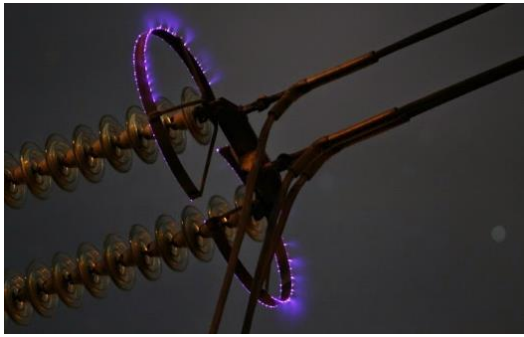
Physical meaning of corona is partial discharge occuring around regions with small curvature radius (sharp edges, corners, needles, ...)



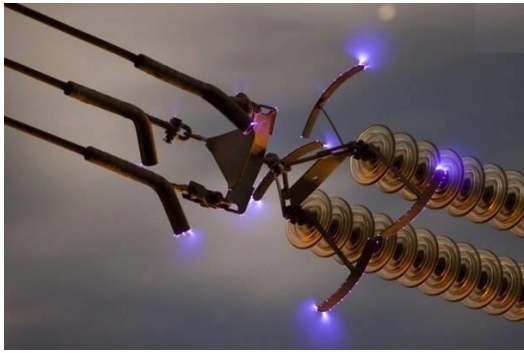
1 DEFINITION

Corona

In a uniform electric field, a gradual increase in voltage across a gap produces a breakdown of the gap in the form of a spark without any preliminary discharges (direct flashover / breakdown without corona dishcarge).



On the other hand, if the field is non-uniform, an increase in voltage will first cause a localised discharge in the gas to appear at points with the highest electric field intensity, namely at sharp points or where the electrodes are curved or on transmission line conductors.



This form of discharge is called a **corona discharge** and can be observed as a bluish luminance.

2 CHARACTERISTIC PROPERTIES

Corona has a specific,

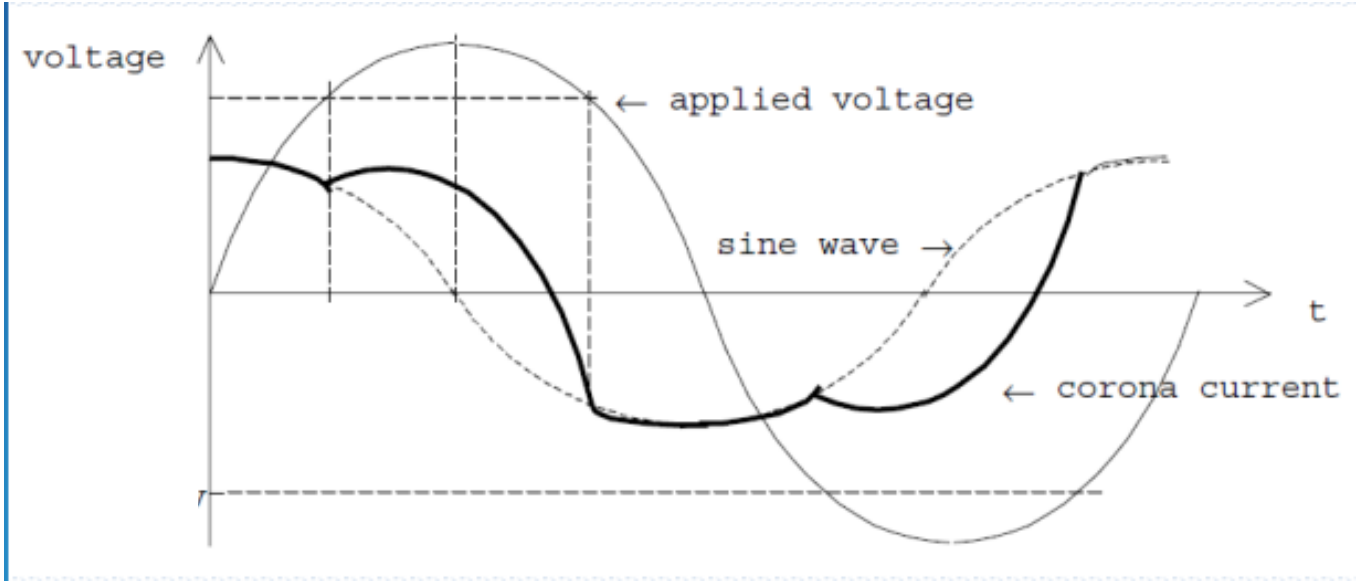
- Light
- Sound (hummin noise)
- Smell (O_3 - ozone gas production)

Corona is more common in transmission lines & equipment of 245 kV and above.

Corona causes power loss, production of high frequency noise, interference with radio and TV, and production of ozone gas.

2 CHARACTERISTIC PROPERTIES

Waveform of a corona current



2 CHARACTERISTIC PROPERTIES

The shunt current in a line is almost purely capacitive under normal conditions, and leads the applied voltage by 90°, and there is no power loss in the line under no-load conditions.

When the applied voltage is increased and corona is formed, the air is rendered conducting and power loss occurs.

The shunt current would no longer be leading the voltage by 90°. Thus the current waveform would consist of two components.

The lossy component would be non-sinusoidal and would occur only when disruptive critical voltage is exceeded in either polarity.

3 MECHANISM OF CORONA

The stress surrounding the conductor is a maximum at the conductor surface itself, and decreases rapidly as the distance from the conductor increases.
Thus, when the stress has been raised to critical value immediately surrounding the conductor, ionisation would commence only in this region and the air in this region would become conducting.
The effect is to increase the effective conductor diameter while the voltage remains constant.

This has **two effects**:

1. An increase in the effective sharpness of the conductor would **reduce the stress** outside this region,
2. This would cause a reduction of the effective spacing between the conductors leading to an **increase in stress**.

Depending on which effect is stronger, the stress at increasing distance can either increase or decrease. If the stress is made to increase, further ionisation would occur and flashover is inevitable.

3 MECHANISM OF CORONA

Under ordinary conditions, the breakdown strength of air can be taken as 30 kV/cm.

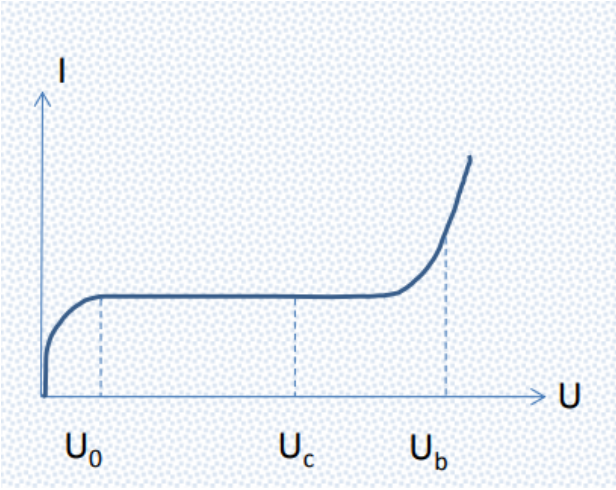
Corona will of course be affected by the physical state of the atmosphere.

In stormy weather, the number of ions present is generally much more than normal, and corona will then be formed at a much lower voltage than in fair weather.

This reduced voltage is generally about 80% of the fair weather voltage.

3 MECHANISM OF CORONA

Current voltage characteristic



POWER LOSS DUE TO CORONA

Power loss due to Corona

The formation of corona is associated with a loss of power.

This loss will have a small effect on the efficieny of the line, but will not be of sufficient importance to have any appreciable effect on the voltage regulation.

The more important effect is the radio interference.

POWER LOSS DUE TO CORONA

Corona Voltage

$$U_c = U_0 \times m \times \delta \times \left(1 + \frac{0.301}{\sqrt{r \times \delta}}\right)$$

Irregularity factor

Corona voltage

Corona inception voltage

curvature radius

Relative air density

m= 1 – 0.8

$$\delta = \frac{p}{p_0} \cdot \frac{T_0}{T} = \frac{p\,[mmHg]}{760[mmHg]} \cdot \frac{273 + 20°C}{273 + \theta°C} = 0.386 \frac{p[mmHg]}{273 + \theta°C}$$

POWER LOSS DUE TO CORONA

For an overhead line

- Single phase:
- $U_c = 21.2 \times m \times r \times \delta \times \left(1 + \frac{0.301}{\sqrt{r \times \delta}}\right) \times \ln \frac{d}{r}$

kVeff

$\frac{30}{\sqrt{2}}$ kV

Electrode gap

- Three phase:
- $U_c = \sqrt{3} \times 21.2 \times m \times r \times \delta \times \left(1 + \frac{0.301}{\sqrt{r \times \delta}}\right) \times \ln \frac{d}{r}$

POWER LOSS DUE TO CORONA

Corona loss (Peek's formula)

$$P_c = \frac{243}{\delta} \times (f + 25) \times \sqrt{\frac{r}{d}} \times (U - U_0)^2 \times 10^{-5} \quad \frac{kW}{km.phase}$$

4 POWER LOSS DUE TO CORONA

Corona can be mainly minimised by,

- increasing the diameter of conductor,
- by bundling the conductor,
- by using corona ring.

END OF THE LESSON

Any questions?

Check out the following videos on Youtube:

Electrical Corona Effect: https://www.youtube.com/watch?v=m-jeQgXAqJA

Flare Corona Discharge: https://www.youtube.com/watch?v=C0Fry6ktu4w

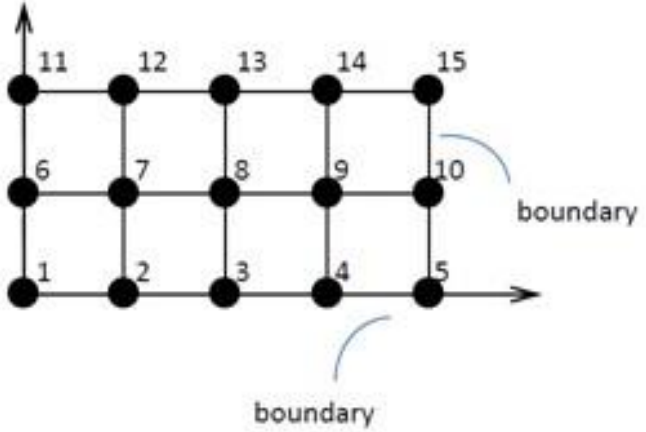
Corona Effect Recorded with Special Camera: https://www.youtube.com/watch?v=5bKcJegKfBo

EE4047 – High Voltage Techniques I

New Topic:

Finite Difference Method

Week 12.1 – Monday Lesson

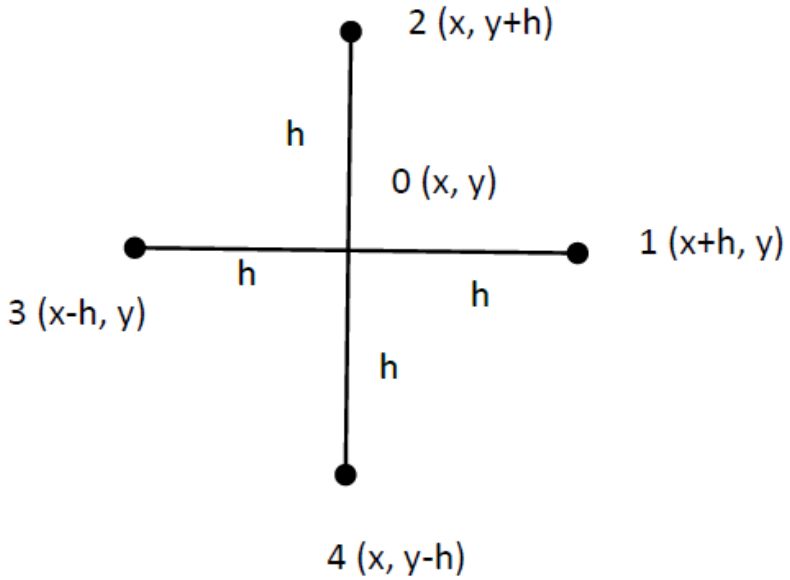


Asst. Prof. Dr. M. Alparslan

16.05.202

FINITE DIFFERENCE METHOD

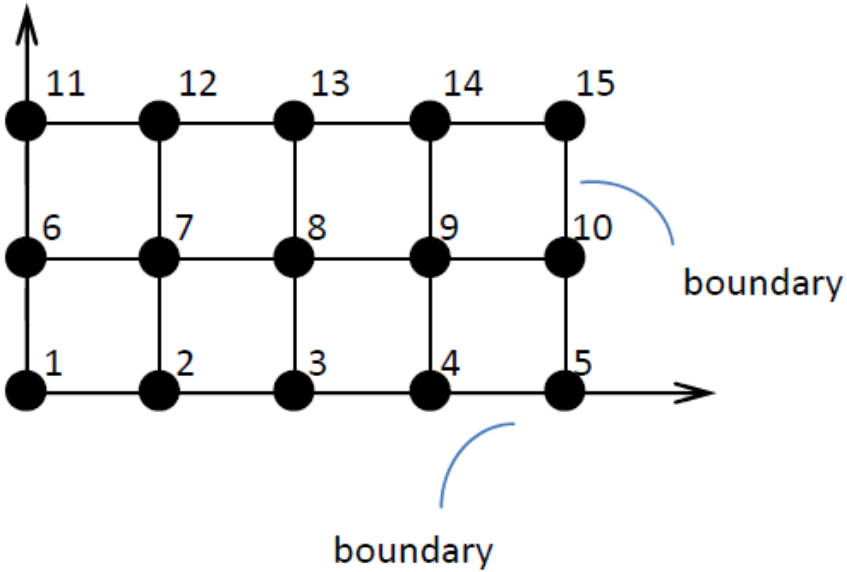
Finite Difference Method



$$V_0 = \frac{1}{4}(V_1 + V_2 + V_3 + V_4)$$

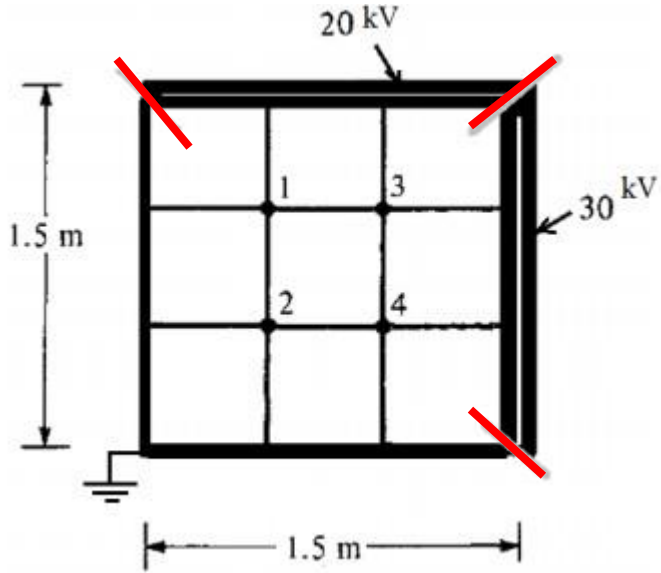
FINITE DIFFERENCE METHOD

Finite Difference Method



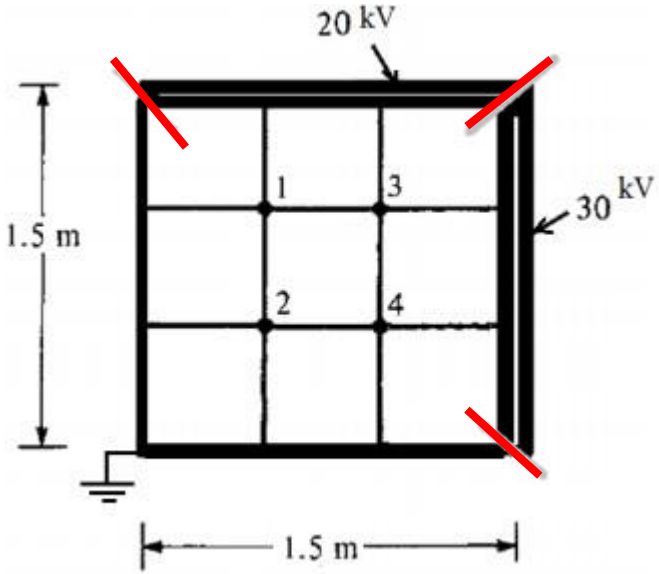
1 FINITE DIFFERENCE METHOD

Example-1: Determine the potential at the free nodes (1, 2, 3 and 4) in the figure using the finite difference method.



1 FINITE DIFFERENCE METHOD

Example-1: Determine the potential at the free nodes (1, 2, 3 and 4) in the figure using the finite difference method.



$$V_1 = \frac{1}{4}(V_2 + V_3 + 0 + 20) \longrightarrow 4V_1 = V_2 + V_3 + 20 \quad (I)$$

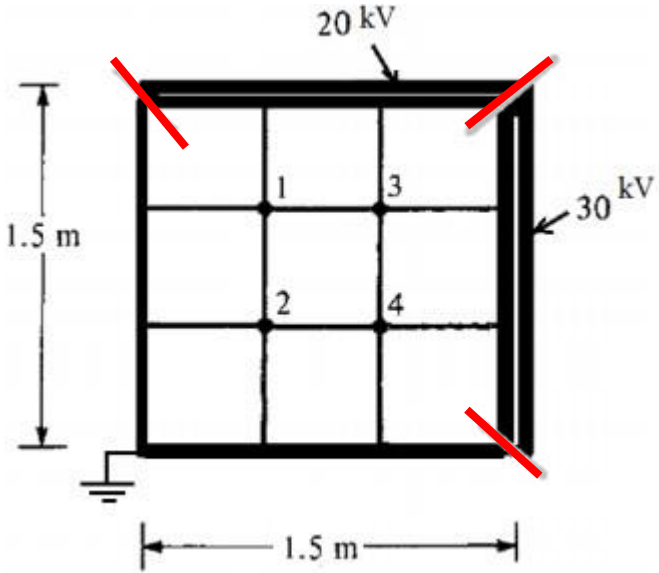
$$V_2 = \frac{1}{4}(V_1 + V_4 + 0 + 0) \longrightarrow 4V_2 = V_1 + V_4 \quad (II)$$

$$V_3 = \frac{1}{4}(V_1 + V_4 + 20 + 30) \longrightarrow 4V_3 = V_1 + V_4 + 50 \quad (III)$$

$$V_4 = \frac{1}{4}(V_2 + V_3 + 30 + 0) \longrightarrow 4V_4 = V_2 + V_3 + 30 \quad (IV)$$

1 FINITE DIFFERENCE METHOD

Example-1: Determine the potential at the free nodes (1, 2, 3 and 4) in the figure using the finite difference method.



(II) & (III) $\quad 4V_3 = 4V_2 + 50 \longrightarrow 4V_2 - 4V_3 = -50$ (V)

(I) & (II) & (IV) $\quad 4V_2 = \frac{1}{4}(V_2 + V_3 + 20) + \frac{1}{4}(V_2 + V_3 + 30)$

$16V_2 = 2V_2 + 2V_3 + 50 \longrightarrow -14V_2 + 2V_3 = -50$ (VI)

1 FINITE DIFFERENCE METHOD

Example-1: Determine the potential at the free nodes (1, 2, 3 and 4) in the figure using the finite difference method.

(V) & (VI)

$$\begin{cases} 4V_2 - 4V_3 = -50 \\ 2\times\ -14V_2 + 2V_3 = -50 \end{cases} \rightarrow \begin{cases} 4V_2 - 4V_3 = -50 \\ -28V_2 + 4V_3 = -100 \end{cases}$$

$$V_2 = \frac{-150}{-24} \Rightarrow V_2 = 6.25 \text{ kV}$$

(V) →

$$V_3 = \frac{-50 - 4V_2}{-4} \rightarrow V_3 = \frac{-50 - 4(6.25)}{-4} \rightarrow V_3 = 18.75 \text{ kV}$$

(I) →

$$V_1 = \frac{V_2 + V_3 + 20}{4} = \frac{6.25 + 18.75 + 20}{4} \rightarrow V_1 = 11.25 \text{ kV}$$

(IV) →

$$V_4 = \frac{V_2 + V_3 + 30}{4} = \frac{6.25 + 18.75 + 30}{4} \rightarrow V_4 = 13.75 \text{ kV}$$

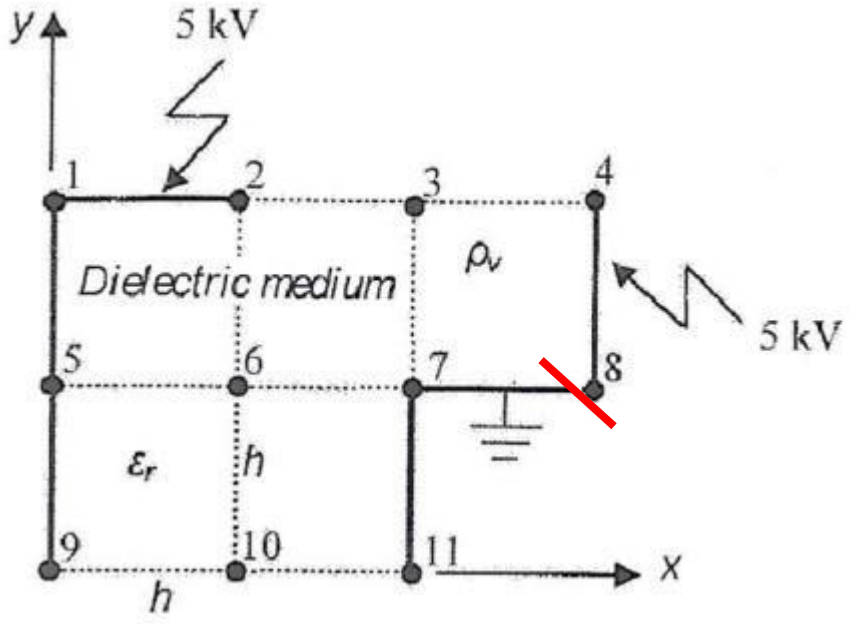
FINITE DIFFERENCE METHOD

Example-2: Determine the potential at the free nodes (3, 6 and 10) in the figure using the finite difference method.

Dirichlet boundary conditions:

$V_1=V_2=V_4=V_5=V_8=V_9=5$ kV, $V_7=V_{11}=0$

Neumann boundary conditions: $(\frac{\partial V}{\partial y})_3 = (\frac{\partial V}{\partial y})_{10} = 0$



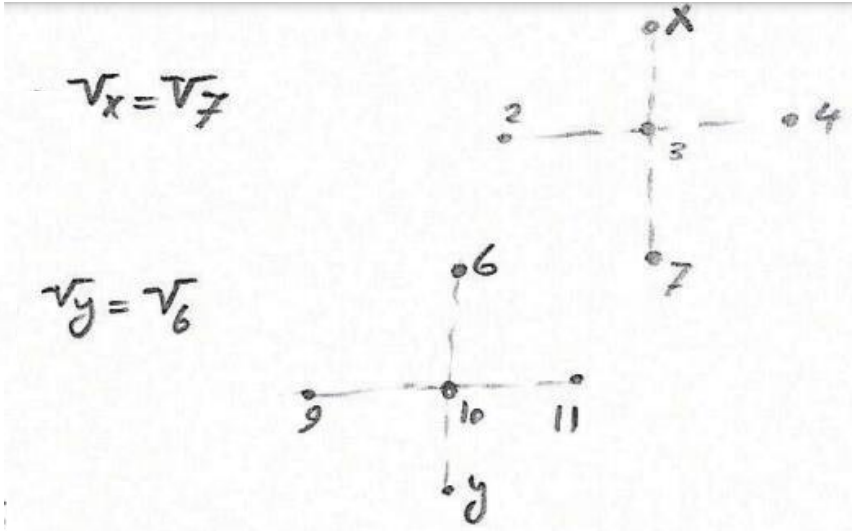
FINITE DIFFERENCE METHOD

Example-2: Determine the potential at the free nodes (3, 6 and 10) in the figure using the finite difference method.

Dirichlet boundary conditions:

$V_1=V_2=V_4=V_5=V_8=V_9=5\ ^{kV}, \quad V_7=V_{11}=0$

Neumann boundary conditions: $(\frac{\partial V}{\partial y})_3 = (\frac{\partial V}{\partial y})_{10} = 0$



$V_x = V_7$

$V_y = V_6$

$V_3 = \frac{1}{4}(V_2 + V_4 + 2V_7) = \frac{1}{4}(5+5+0) \Rightarrow$

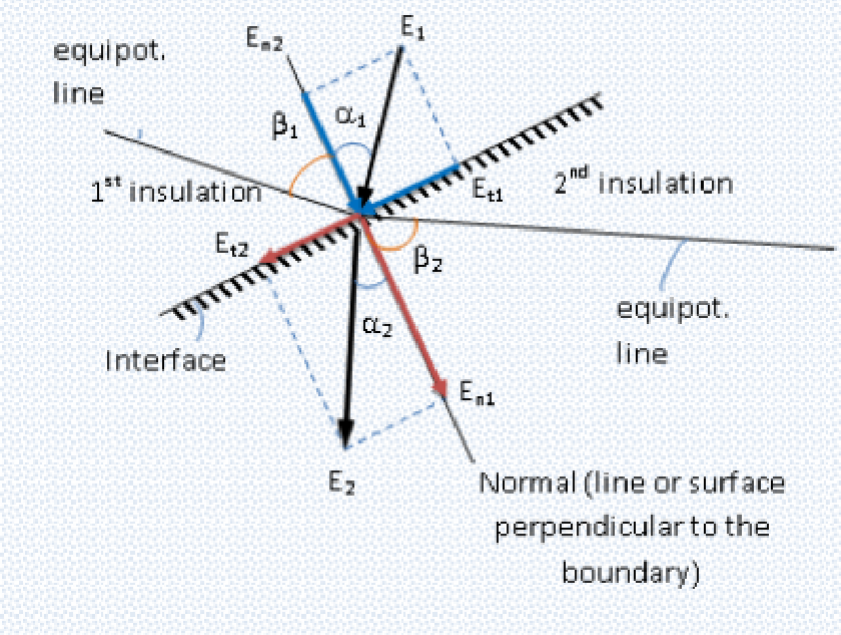
$V_3 = 2.5\ ^{kV}$

EE4047 – High Voltage Techniques I

New Topic:

Refraction of Electric Fields

Week 12.1 – Monday Lesson



Asst. Prof. Dr. M. Alparslan

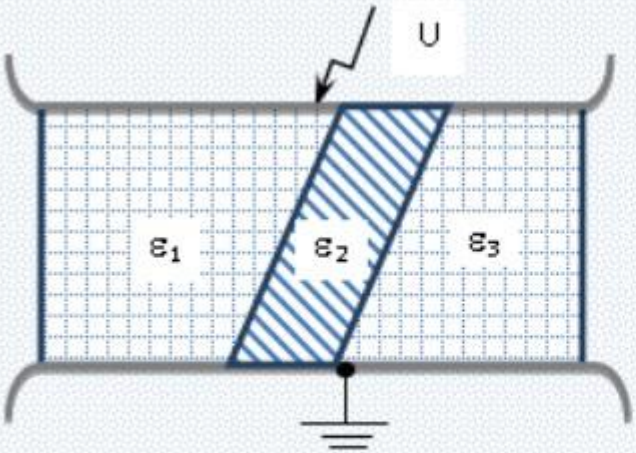
17.05.202

OUTLINE

1 Introduction

2 Refraction

Non-parallel configuration

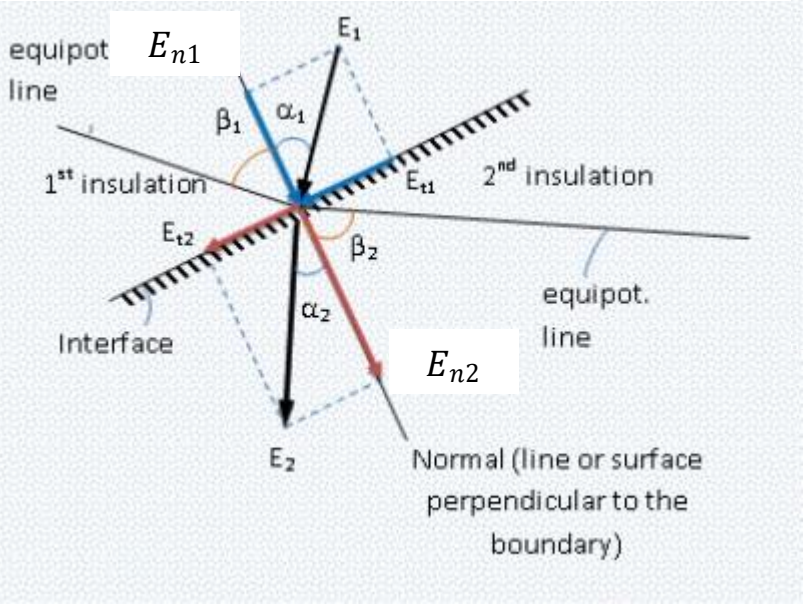


What happens to electric field if the plates are NOT parallel?

In order to evaluate that, we have to know the behaviour of field lines at boundary surfaces.

2 REFRACTION

Refraction of electric field lines and equipotential lines



- α_1, α_2: Refraction angles of electric field lines.
- β_1, β_2: Refraction angles of equipotential lines

(REMEMBER: always perpendicular to the field lines!!!)

- E_{t1}, E_{t2}: tangent component of the electric fields E_1 and E_2
- E_{n1}, E_{n2}: normal (perpendicular) component of the electric fields E_1 and E_2

$$E_{t1} = E_{t2}$$

END OF THE LESSON

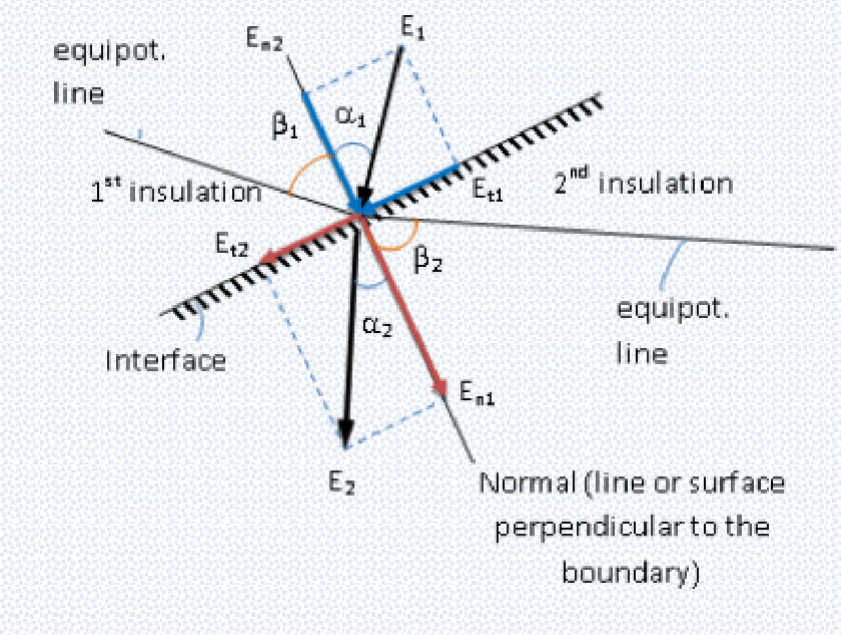
Any questions?

EE4047 – High Voltage Techniques I

New Topic:

Refraction of Electric Fields

Week 13.1 – Monday Lesson



Asst. Prof. Dr. M. Alparslan

29.05.2023

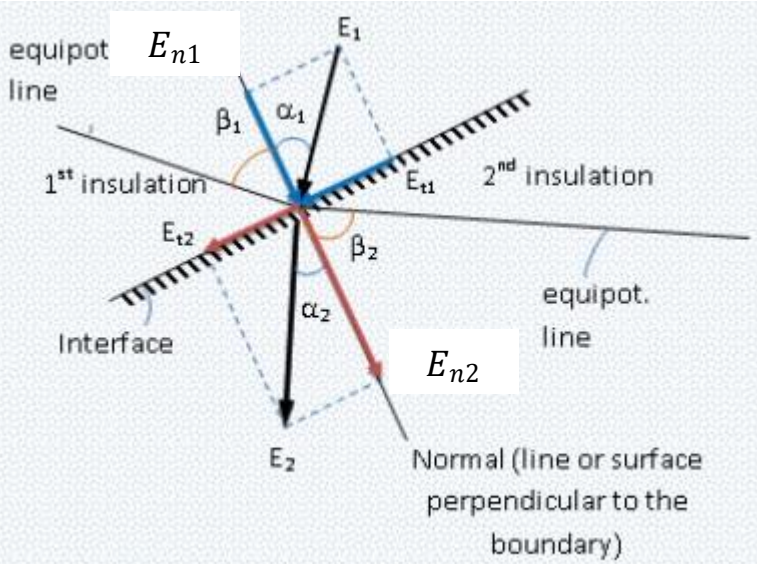
OUTLINE

1 Introduction

2 Refraction

2 REFRACTION

Refraction of electric field lines and equipotential lines



- α_1, α_2: Refraction angles of electric field lines.
- β_1, β_2: Refraction angles of equipotential lines

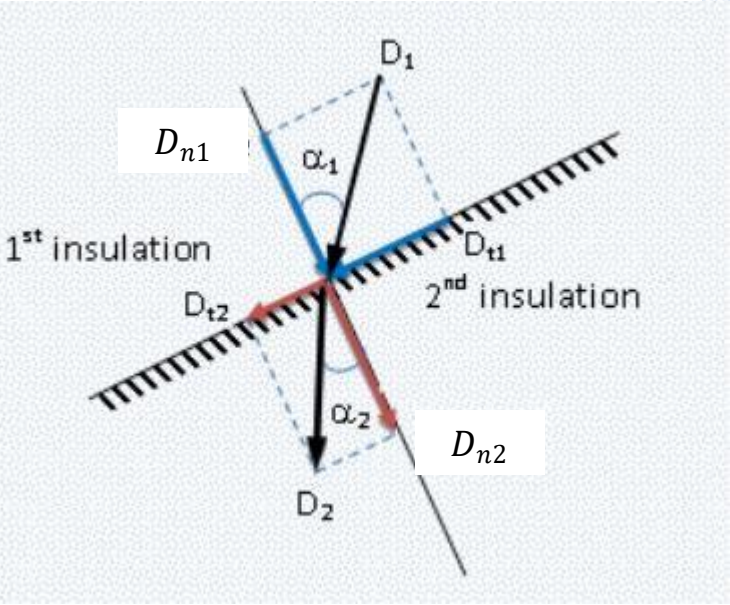
(REMEMBER: always perpendicular to the field lines!!!)

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- E_{n1}, E_{n2}: normal (perpendicular) component of the electric fields E_1 and E_2

$$E_{t1} = E_{t2}$$

2 REFRACTION

Refraction of electric field lines and equipotential lines

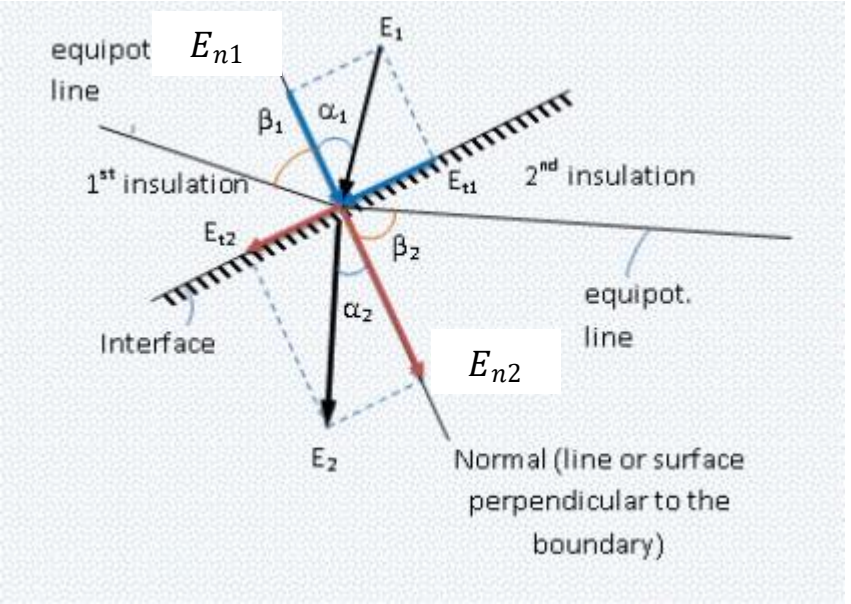


Similar to the electric field lines, displacement lines are refracted at the interface so that;

$$D_{n1} = D_{n2}$$

2 REFRACTION

Simple trigonometric concepts



For electric field vector components:

$$E_{t1} = E_1 \sin \alpha_1$$
$$E_{t2} = E_2 \sin \alpha_2$$

For displacement vector components:

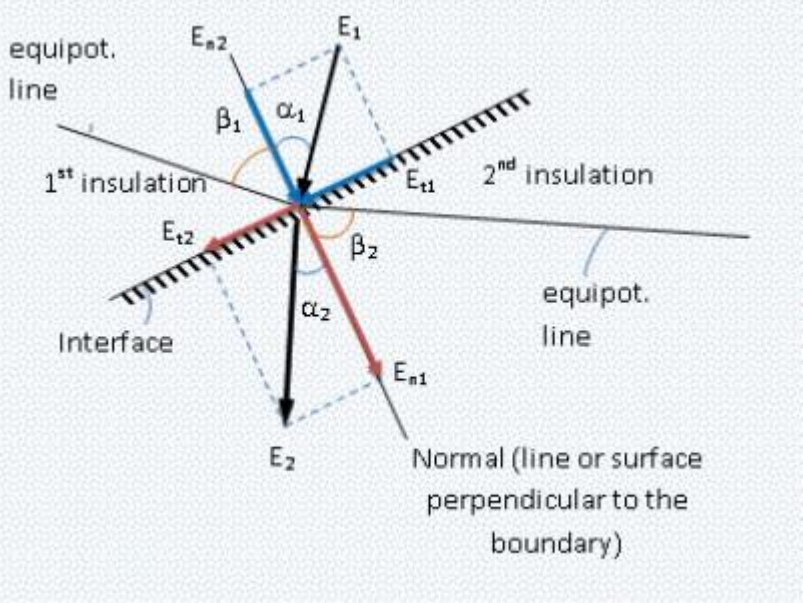
$$D_{n1} = D_1 \cos \alpha_1 = \varepsilon_1 E_1 \cos \alpha_1$$
$$D_{n2} = D_2 \cos \alpha_2 = \varepsilon_2 E_2 \cos \alpha_2$$

Because $E_{t1} = E_{t2}$ and $D_{n1} = D_{n2}$; we can write

$$\frac{E_{t1} = E_{t2}}{D_{n1} = D_{n2}}.$$

2 REFRACTION

Refraction of equipotential lines



$$\beta_1 = 90 - \alpha_1$$
$$\beta_2 = 90 - \alpha_2$$

refraction angles of equipotential lines.

$$\frac{\tan(90-\beta_1)}{\tan(90-\beta_2)} = \frac{\varepsilon_1}{\varepsilon_2}$$

2 REFRACTION

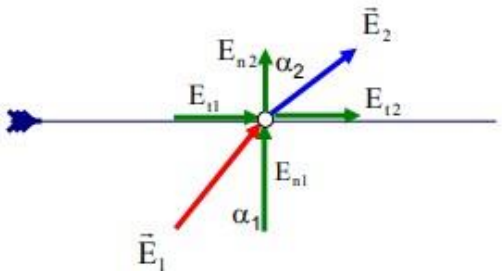
$$E_{t1} = E_{t2}$$

$$\frac{D_{t1}}{D_{t2}} = \frac{\varepsilon_1}{\varepsilon_2}$$

$$\vec{D} = \varepsilon.\vec{E}$$

$$\frac{E_{n1}}{E_{n2}} = \frac{\varepsilon_2}{\varepsilon_1}$$

$$D_{n1} = D_{n2}$$



2 REFRACTION

Refraction of equipotential lines

$$E_{t1} = E_{t2} \Rightarrow |E_1|.\sin\alpha_1 = |E_2|.\sin\alpha_2 \qquad \frac{\tan\alpha_1}{\tan\alpha_2} = \frac{\varepsilon_1}{\varepsilon_2} \qquad \alpha_1 + \beta_1 = 90°$$

$$D_{n1} = D_{n2} \Rightarrow |D_1|.\cos\alpha_1 = |D_2|.\cos\alpha_2 \qquad \frac{\tan\beta_1}{\tan\beta_2} = \frac{\varepsilon_2}{\varepsilon_1} \qquad \alpha_2 + \beta_2 = 90°$$

$$|E_2| = \sqrt{E^2_{t2} + E^2_{n2}} = \sqrt{E^2_{t1} + (\frac{\varepsilon_1}{\varepsilon_2})^2.E^2_{n1}}$$

2 REFRACTION

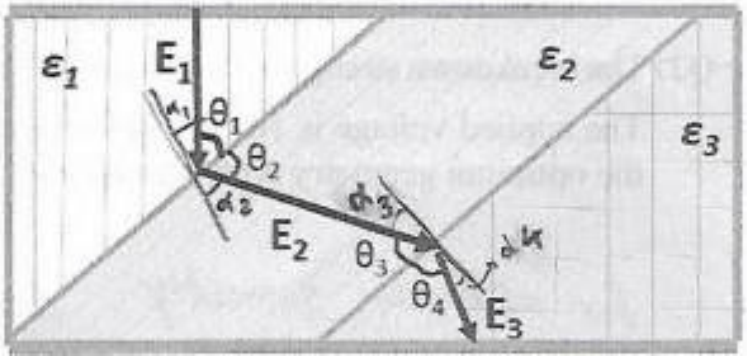
Refraction of equipotential lines

Q4) In the following insulation system, there is a discharge in the 3rd insulator. The relative permittivity of the third insulator is $\varepsilon_{r3} = 1.2$ and its electric field strength is $E_{b3} = 100$ kV/cm. Suppose $\theta_1 = 45^\circ$, $\theta_2 = \theta_3 = 60^\circ$ and $\theta_4 = 75^\circ$. Determine:

a) The minimum values of E_1 and E_2.

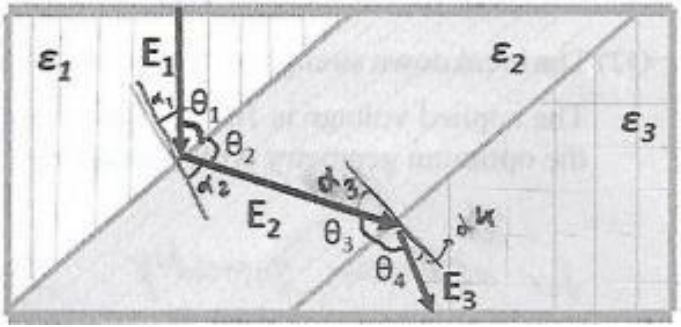
b) The relative permittivities of ε_{r1} and ε_{r2}.

($\varepsilon_0 = 8.85 \times 10^{-12}$)



2 REFRACTION

Refraction of equipotential lines



(a)

If breakdown happend in 3rd layer $\Rightarrow$ E_3 is minimum $100 \frac{kV}{cm}$

$$E_3 \sin\alpha_4 = E_2 \sin\alpha_3 \rightarrow 100 \sin(90^\circ - 75^\circ) = E_2 \sin(90^\circ - 60^\circ)$$

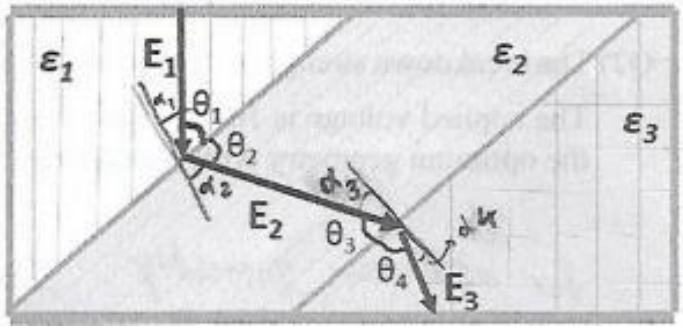
$$E_2 = \frac{100 \sin(15^\circ)}{\sin(30^\circ)} \rightarrow E_{2\,Minimum} = 51.76 \frac{kV}{cm}$$

$$E_2 \sin\alpha_2 = E_1 \sin\alpha_1 \longrightarrow 51.76 \sin(90^\circ - 60^\circ) = E_1 \sin(90^\circ - 45^\circ)$$

$$E_1 = \frac{51.76 \sin(30^\circ)}{\sin(45^\circ)} \Rightarrow E_{1\,Minimum} = 36.60 \frac{kV}{cm}$$

2 REFRACTION

Refraction of equipotential lines



$$\frac{tan\alpha_2}{tan\alpha_3} = \frac{\varepsilon_2}{\varepsilon_3}$$

$$\frac{tan\alpha_2}{tan\alpha_3} = \frac{\varepsilon_{r2}}{\varepsilon_{r3}}$$

$$\frac{\tan(90-\theta_2)}{tan(90-\theta_3)} = \frac{\varepsilon_{r2}}{\varepsilon_{r3}}$$

(b)

$$\frac{\varepsilon_2}{\varepsilon_3} = \frac{tg\,\beta_2}{tg\,\beta_2} \rightarrow \frac{\varepsilon_0 \varepsilon_{r2}}{\varepsilon_0 (1.2)} = \frac{tg\,75^\circ}{tg\,60^\circ} \rightarrow \varepsilon_{r2} = 2.856$$

$$\frac{\varepsilon_1}{\varepsilon_2} = \frac{tg\,\beta_2}{tg\,\beta_1} \rightarrow \frac{\varepsilon_0 \varepsilon_{r1}}{\varepsilon_0 (2.856)} = \frac{tg\,60^\circ}{tg\,45^\circ} \rightarrow \varepsilon_{r1} = 4.647$$

2 REFRACTION

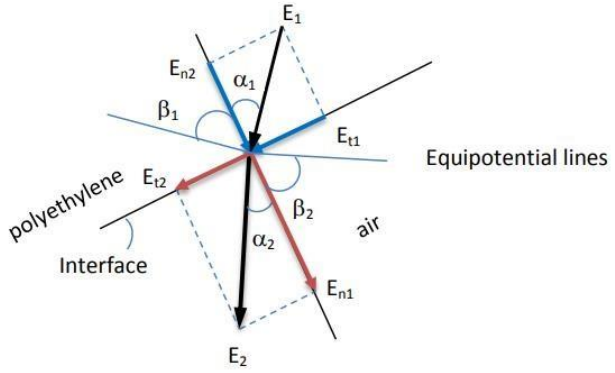
Refraction of equipotential lines

Q3. An insulation environment consists of polyethylene ($\varepsilon_{r1} = 1$) and air ($\varepsilon_{r2} = 2.3$). Electric field vector comes to the interface with an angle of 55° through polyethylene.

a. Determine the refraction angles of electric field lines and equipotential lines.

b. If electric field strength in polyethylene is $E_1 = 90$ kV/cm, determine electric field strength in air, E_2; normal and tangential components of E_1 and E_2.

c. Does air have a breakdown considering breakdown strength of air is 30 kV/cm?



2 REFRACTION

Refraction of equipotential lines

a. $E_1 = 90$ kV/cm

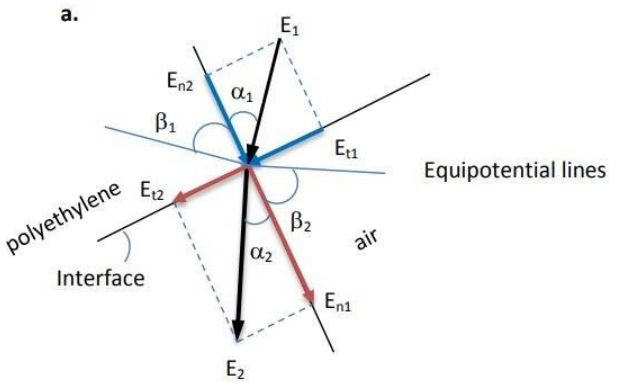
$\alpha_1 = 90 - 55 = 35°$

$\beta_1 = 90 - \alpha_1 = 55°$

$$\frac{\tan\alpha_1}{\tan\alpha_2} = \frac{\varepsilon_1}{\varepsilon_2} \rightarrow \tan\alpha_2 = \frac{1}{2.3}\tan 35°$$

$\alpha_2 = 16.932°$

$\beta_2 = 90 - \alpha_2 = 73.068°$



2 REFRACTION

Refraction of equipotential lines

b.

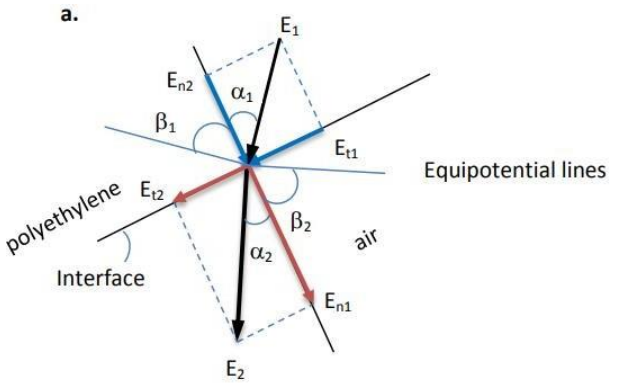
$$E_{t1} = E_1 \sin\alpha_1 = 90\frac{kV}{cm}\sin 35^\circ = 51.62\ \frac{kV}{cm}$$

$$E_{n1} = E_1\cos\alpha_1 = 90\frac{kV}{cm}\cos 35^\circ = 73.722\ \frac{kV}{cm}$$

$$E_{t2} = E_{t1} = E_2 \sin\alpha_2 \rightarrow E_2 = \frac{E_{t2}}{\sin\alpha_2} = \frac{51.62\ \frac{kV}{cm}}{\sin 16.932^\circ} = 177.244\frac{kV}{cm}$$

$$E_{n2} = E_2\cos\alpha_2 = 177.244\frac{kV}{cm}\cos 16.932^\circ{}^\circ = 169.56\ \frac{kV}{cm}$$

c. $E_{n2} = 169.56\ \frac{kV}{cm} > E_{b2} = 30\frac{kV}{cm} \rightarrow breakdown\ in\ air!$



END OF THE LESSON

Any questions?

Check out the following video on Youtube:

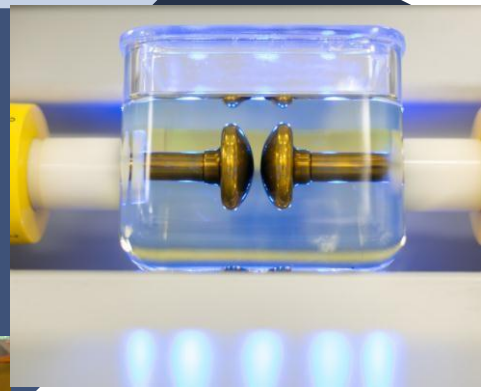
Refraction of electric field lines: https://www.youtube.com/watch?v=A4VpnsKv5QI

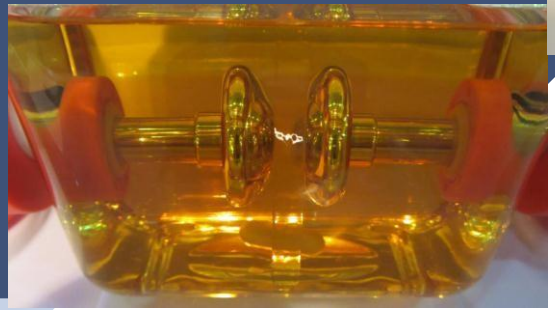
EE4047 – High Voltage Techniques I

New Topic:

Breakdown in Liquids

Week 14.2 – Monday Lesson





Asst. Prof. Dr. M. Alparslan

08.06.2023

OUTLINE

1. Liquid Dielectrics
2. Breakdown in liquids
3. Factors affecting breakdown in liquids
4. Breakdown types in liquid dielectrics

LIQUID DIELECTRICS

Liquid dielectrics

- Insulation oil
- Transformer oil
- CCl_4 (carbon tetrachloride)

Where to use them?

- Insulation
- Cooling (transfers heat through natural flow in the oil: cold oil replaces heated oil inside a transformer or breaker or other)
- Protection (extinguishing arcs, protecting the metalic surfaces from humidity, dirt and rust)

1 LIQUID DIELECTRICS

Mineral oils are processed products of petroleum. They have Fluorine (F) and Clorine (Cl). These oils may become flamable at high temperatures.

2 BREAKDOWN IN LIQUIDS

- In highly purified liquid dielectrics, breakdown is controlled by phonomena similar to those for gasses and the electric breakdown strength is high (of the order of 1 MV/cm).

- Unfortunately, liquids are easily contaminated and may contain solids, other liquids in suspension and dissolved gasses which are conductive or have less breakdown strength compared to the liquid dielectric they are in.

- Contamination considerably lowers breakdown strength of liquid dielectrics.

- The impact of these impurities is relatively small for short duration pulses (10 s).

- However, if the voltage is applied continuously, the solid impurities line up at right angles to equipotentials and distort the field, so that breakdown occurs at relatively low voltage.

- The line up of particles is a fairly slow process and is unlikely to affect the strength on voltages lasting for less than 1 ms.

2 BREAKDOWN IN LIQUIDS

- Due to the tendency to become contaminated, liquid dielectrics are not usually used alone above 100 kV/cm in continuously energised equipment.

- They are used at much higher stresses (up to 1 MV/cm) in conjunction with solids, which can be made to act as barriers, preventing the line-up of solid impurities and localising of any bubbles which may form.

- The main function of the liquid, when it is used together with solid dielectrics, is to fill up the gaps.

- Breakdown strength depends more on the nature of impurities a liquid dielectric contains than on the nature of the liquid itself.

- These impurities lead to the breakdown of liquid dielectrics below their intrinsic strength.

3 FACTORS AFFECTING BREAKDOWN IN LIQUIDS

1. Temperature (if increases, breakdown voltage **decreases**)
2. Humidity (if increases, breakdown voltage **decreases**)
3. Pressure (if increases, breakdown voltage **increases** too)
4. Solid impurities:dust, dirt, fiber particles (breakdown voltage **decreases**)
5. Gas bubbles (breakdown voltage **decreases**)
6. Electrode geometry, material, surface and polarity (iron has lower breakdown strength than silver)
7. Voltage type, application duration (duration reduces breakdown voltage, DC breakdown voltage is lower than AC)
8. Frequency

FACTORS AFFECTING BREAKDOWN IN LIQUIDS

Impurities

-Impurities which have a breakdown strength lower than the liquid itself (such as bubbles of gas). Breakdown of the impurities may trigger off the total breakdown of the liquid.

-Impurities which are unstable in the electric field (such as globules of water). Instability of the impurity can result in a low resistance bridge across the electrodes, leading to total breakdown.

- Impurities which result in local enhancement of electric field in a liquid (such as conducting particles). The enhanced field may cause local breakdown and initiate complete breakdown.

BREAKDOWN TYPES IN LIQUID DIELECTRICS

1. **Electronic breakdown** (based on Townsend Theory in gases, happens in pure liquid dielectrics)

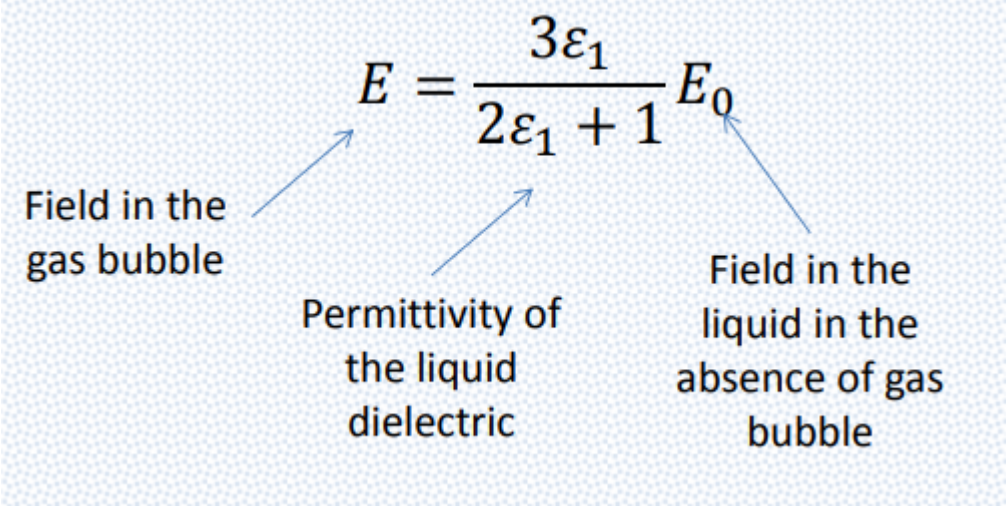
2. **Electromechanical breakdown** (there could be no particles in the liquid, but there may be air bubbles or gas globules, which cause this type of breakdown)

3. **Thermal breakdown** (the particles in the impurified liquids form a bridge and causes breakdown)

BREAKDOWN TYPES IN LIQUID DIELECTRICS

Breakdown due to gaseous inclusions

Gas or vapour bubbles may exist in impure liquid dielectrics, either formed from dissolved gasses, temperature and pressure variations, or other causes.



BREAKDOWN TYPES IN LIQUID DIELECTRICS

Breakdown due to gaseous inclusions

The electrostatic forces on the bubble cause it to get elongated in the direction of the electric field.

The elongation continues, when sufficient electric field is applied, and at a critical length the gas inside the bubble (which has a relatively lower breakdown strength) breaks down.

This discharge causes decomposition of the liquid molecules and leads to total breakdown.

BREAKDOWN TYPES IN LIQUID DIELECTRICS

Breakdown due to liquid globule

If an insulating liquid contains in suspension a globule of another liquid, then breakdown can result from instability of the globule in the electric field.

$$E = \frac{3\varepsilon_1}{2\varepsilon_1 + \varepsilon_2} E_0$$

- E: Field in the globule
- ε_1: Permittivity of the liquid dielectric
- ε_2: Permittivity of the spherical globule
- E_0: Field in the liquid in the absence of the globule

BREAKDOWN TYPES IN LIQUID DIELECTRICS

Breakdown due to solid particles

In commercial liquids, solid impurities cannot be avoided and will be present as fibres or as dispersed solid particles.

If the impurity is considered to be a spherical particle of permittivity ε_2 and is present in a liquid dielectric of permittivity ε_1, it will experience a force

$$F = \frac{\varepsilon_2 - \varepsilon_1}{\varepsilon_2 + 2\varepsilon_1} \Delta E^2$$

BREAKDOWN TYPES IN LIQUID DIELECTRICS

Breakdown due to solid particles

Generally $\varepsilon_2 > \varepsilon_1$, so that the force would move the particle towards the regions of stronger field.

Particles will continue to move in this way and will line up in the direction of the field.

A stable chain of particles would be which at a critical length may cause breakdown.

BREAKDOWN TYPES IN LIQUID DIELECTRICS

The particles can be taken out of the oil by filtering the oil.

Humidity and water bubbles can be eliminated by drying the oil.

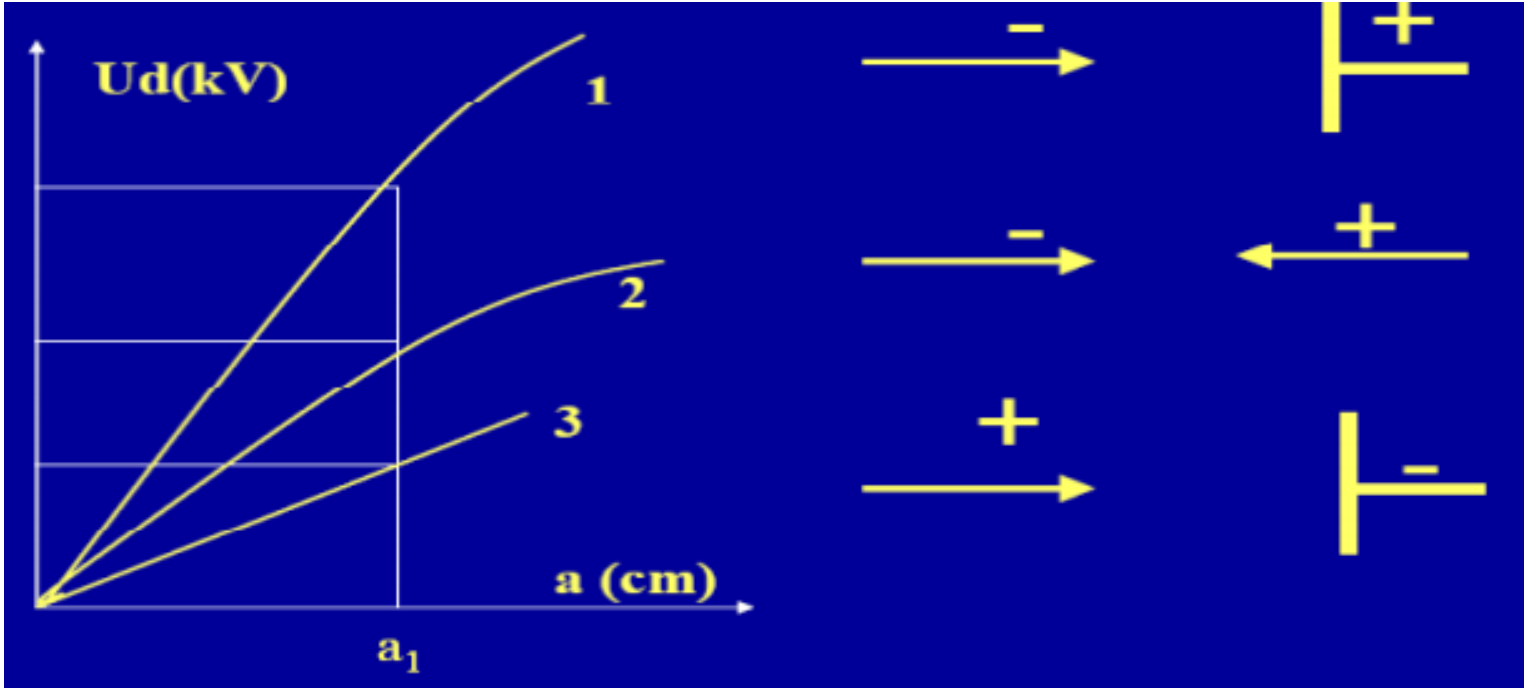
BREAKDOWN TYPES IN LIQUID DIELECTRICS

Desired dielectric oil properties

- Purity (mineral oil with no addition of other materials, also no water-gas-solid particles)
- High flamability point (becomes flamable at a very high temperature)
- High dielectric strength ($E_b > 200\ kV/cm$)
- High liquidity at low temperatures (should not become considerable less liquid due to cold)
- Enough liquidity (low viscosity) at any temperature to extinguish arc
- Should not be very acidic

BREAKDOWN TYPES IN LIQUID DIELECTRICS

Breakdown due electrode polarity and geometry



END OF THE LESSON

Any questions?

Check out the following videos on Youtube:

Breakdown voltage testing of transformer oil: https://www.youtube.com/watch?v=nZAg_A1tb0o

Animation of transformer oil testing: https://www.youtube.com/watch?v=BnrARAP2btk

Transformer oil test on site training: https://www.youtube.com/watch?v=YTp-SjvaA5Q

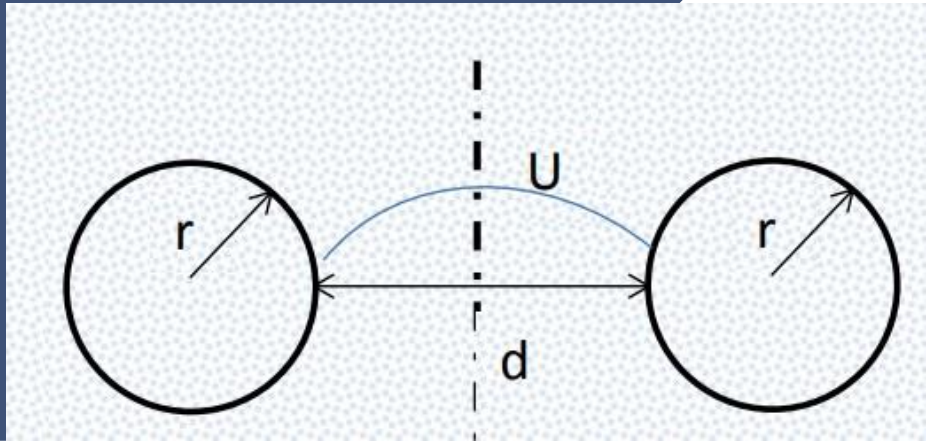
Oil testing equipment video: https://www.youtube.com/watch?v=WxVKADaLyXk

EE4047 – High Voltage Techniques I

New Topic:

Approximated Electric Field Calculations

Week 14.2 – Monday Lesson



Asst. Prof. Dr. M. Alparslan

08.06.2023

OUTLINE

1 Plane-Sphere Electrode System

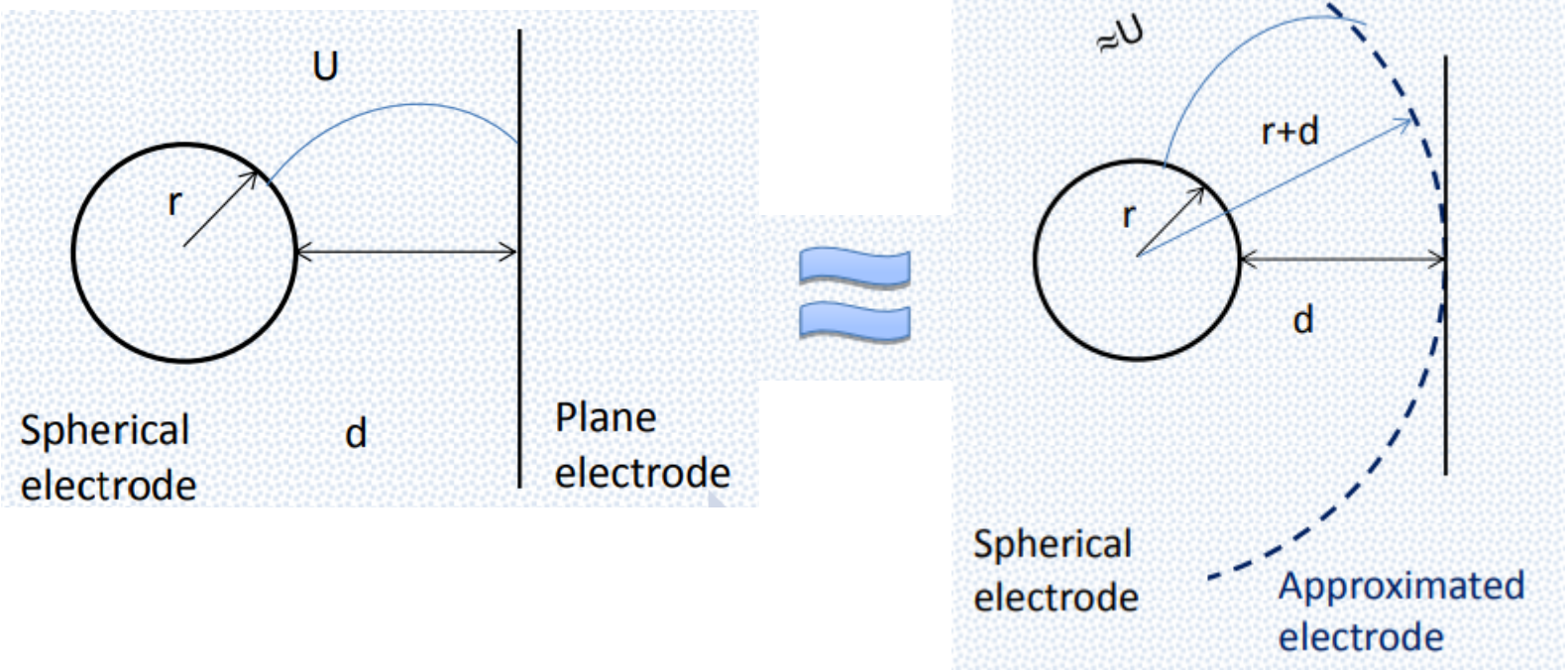
2 Sphere-Sphere Electrode System

3 Plane-Cylinder Electrode System

4 Cylinder-Cylinder Electrode System

1 PLANE-SPHERE ELECTRODE SYSTEM

Plane-sphere electrode system



PLANE-SPHERE ELECTRODE SYSTEM

Plane-sphere electrode system

$$E_{max} = \frac{U}{\frac{r_1}{r_2}(r_2 - r_1)}$$

$r_1 = r$, $r_2 = r + d$, $U = U$

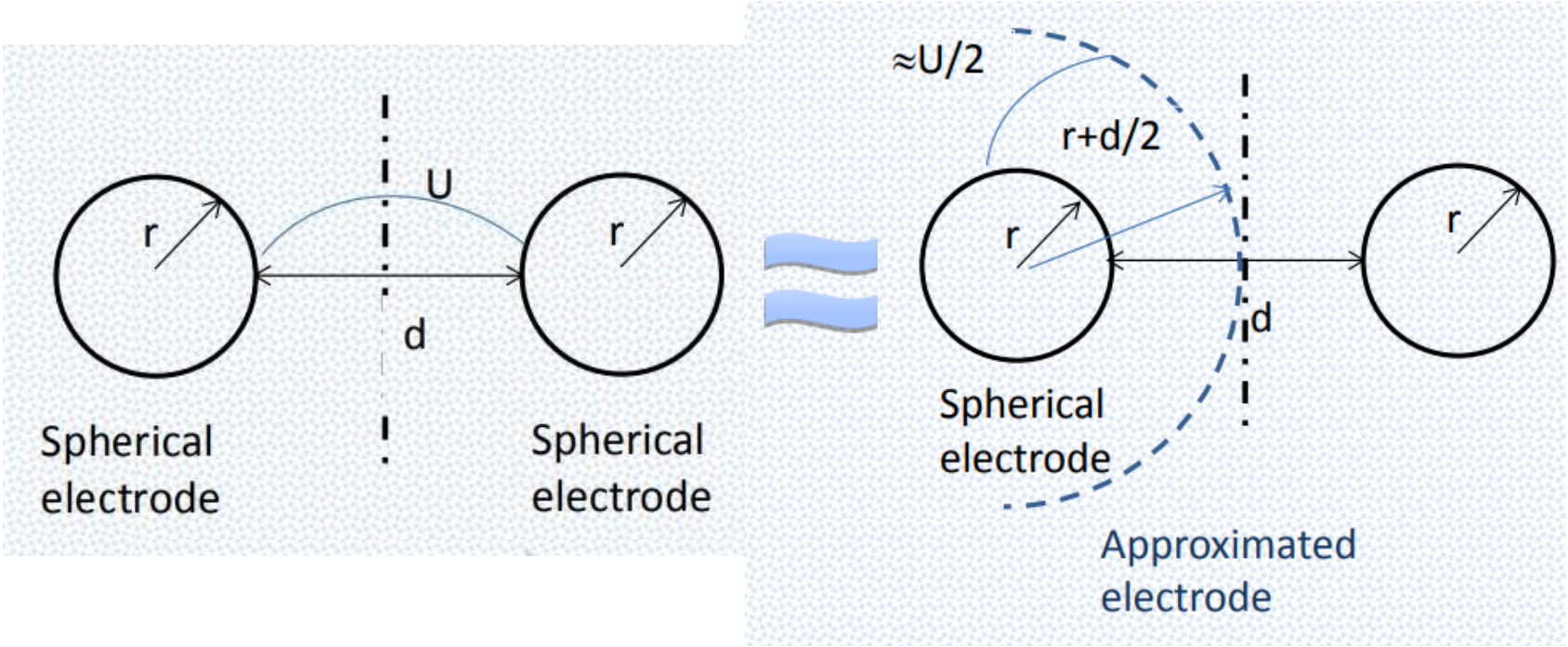
$$E_{max} = \frac{U}{\frac{r}{r + d}(r + d - r)}$$

$$E_{max} = \boxed{k}\frac{U}{d}.\frac{r + d}{r}$$

k ≅ 0.85 - 0.95

SPHERE-SPHERE ELECTRODE SYSTEM

Sphere-sphere electrode system



SPHERE-SPHERE ELECTRODE SYSTEM

Sphere-sphere electrode system

$$E_{max} = \frac{U}{\frac{r_1}{r_2}(r_2 - r_1)}$$

$r_1 = r,\ r_2 = r + d/2,\ U = U/2$

$$E_{max} = \frac{U/2}{\frac{r}{r + d/2}(r + d/2 - r)}$$

$$E_{max} = \boxed{k}\frac{U}{d}.\frac{r + d/2}{r}$$

$k \cong 0.85 - 0.95$

2 SPHERE-SPHERE ELECTRODE SYSTEM

Example:

In a sphere-sphere (identical spheres) electrode system, the breakdown voltage is 13 kV. The correction factor in approximated field calculation is k=0.95 and breakdown strength of air is supposed to be E_b =30 $^{kV/cm}$. Determine radii of the spheres (r) and electrode separation (d) when r = 10 d.

2 SPHERE-SPHERE ELECTRODE SYSTEM

Example:

In a sphere-sphere (identical spheres) electrode system, the breakdown voltage is 13 kV. The correction factor in approximated field calculation is k=0.95 and breakdown strength of air is supposed to be E_b =30 kV/cm. Determine radii of the spheres (r) and electrode separation (d) when r = 10 d.

$$E_b = K \frac{U_b/2}{\frac{r}{r+d/2}(r+d/2-r)} = K \frac{U_b}{2} \frac{2r+d}{dr} \quad \& \quad r=10d$$

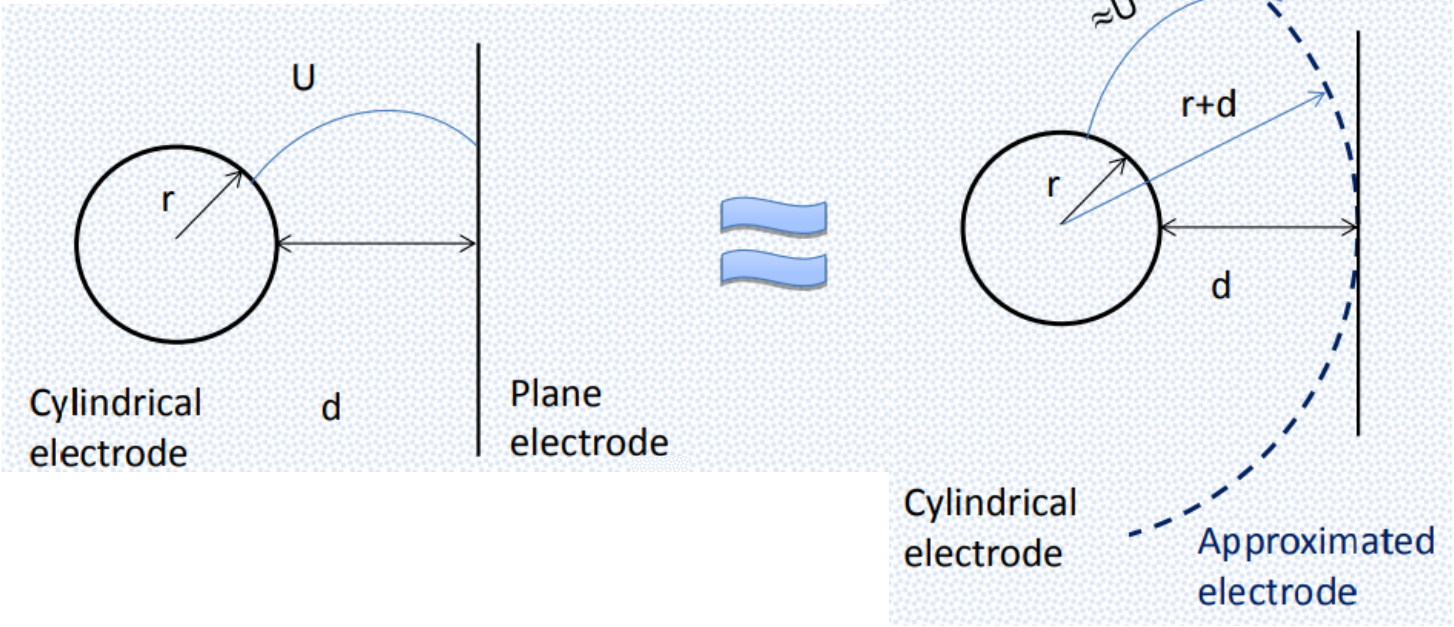


$$30 \frac{kV}{cm} = 0.95 \frac{13}{2} \frac{2(10d)+d}{d(10d)} = 6.175 \frac{21d}{10d^2} = \frac{12.9675}{d} \rightarrow d = 0.43225\,cm$$

$$r = 10d = 4.3225$$

PLANE-CYLINDER ELECTRODE SYSTEM

Plane-cylinder electrode system



PLANE-CYLINDER ELECTRODE SYSTEM

Plane-cylinder electrode system

$$E_{max} = \frac{U}{r_1 \ln \frac{r_2}{r_1}}$$

$r_1 = r,\ r_2 = r + d,\ U = U$

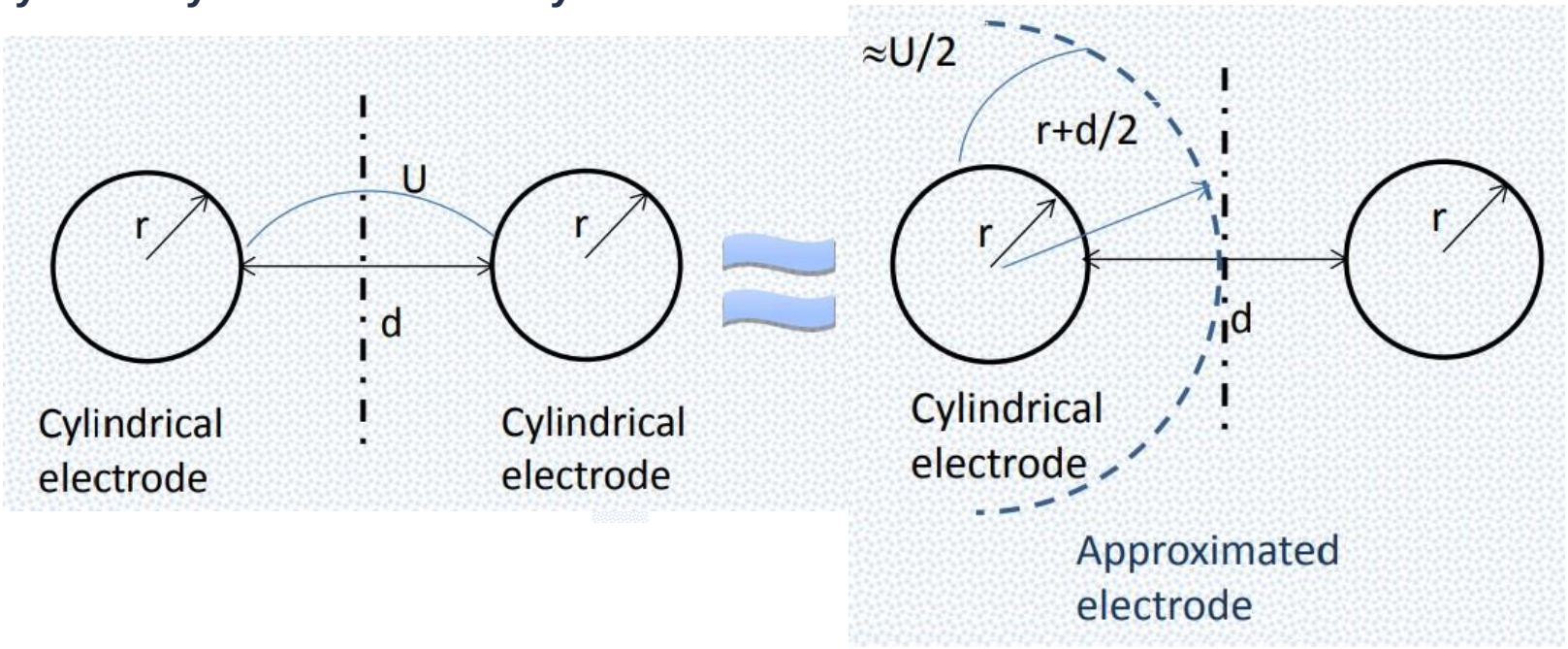
$$E_{max} = \frac{U}{r \ln \frac{r + d}{r}}$$

$$E_{max} = k \frac{U}{r \ln \frac{r + d}{r}}$$

$k \cong 0.85 - 0.95$

CYLINDER-CYLINDER ELECTRODE SYSTEM

Cylinder-cylinder electrode system



CYLINDER-CYLINDER ELECTRODE SYSTEM

Cylinder-cylinder electrode system

$$E_{max} = \frac{U}{r_1 \ln \frac{r_2}{r_1}}$$

$r_1 = r,\ r_2 = r + d/2,\ U = U/2$

$$E_{max} = \frac{U/2}{r \ln \frac{r + d/2}{r}}$$

$$E_{max} = k \frac{U}{2r \ln \frac{r + d/2}{r}}$$

$k \cong 0.85 - 0.95$

CYLINDER-CYLINDER ELECTRODE SYSTEM

Example:

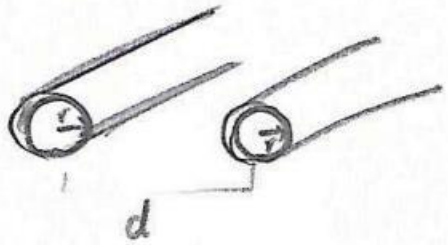
In a three phase 380 ^{kV} transmission line, diameter of the conductors are 32 ^{mm}. The breakdown strength of air is 30 $^{kV/cm}$ and the correction factor is k=0.85. Using approximated maximum field calculation, determine the minimum distance between two phases.

CYLINDER-CYLINDER ELECTRODE SYSTEM

Example:

In a three phase 380 ^{kV} transmission line, diameter of the conductors are 32 ^{mm}. The breakdown strength of air is 30 $^{kV/cm}$ and the correction factor is k=0.85. Using approximated maximum field calculation, determine the minimum distance between two phases.

$r = \frac{32}{2} = 16^{mm} = 1.6^{cm}$



$E_{MAX} = k \frac{U/2}{r \ln\left(\frac{r+d/2}{r}\right)} \rightarrow 30 \frac{kV}{cm} = 0.85 \frac{380/2}{1.6 \ln\left(\frac{1.6+\frac{d}{2}}{1.6}\right)}$

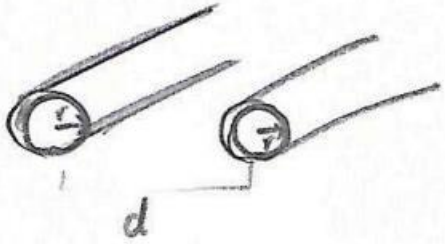
$\rightarrow \ln\left(\frac{1.6+d/2}{1.6}\right) = \frac{0.85 \times 190}{1.6 \times 30} = 3.365$

CYLINDER-CYLINDER ELECTRODE SYSTEM

Example:

In a three phase 380 ^{kV} transmission line, diameter of the conductors are 32 ^{mm}. The breakdown strength of air is 30 $^{kV/cm}$ and the correction factor is k=0.85. Using approximated maximum field calculation, determine the minimum distance between two phases.

$r = \frac{32}{2} = 16^{mm} = 1.6^{cm}$



$$\frac{1.6 + d/2}{1.6} = e^{3.365} = 28.92 \rightarrow 1.6 + \frac{d}{2} = 46.272$$

$$d = 89.34^{cm}$$

END OF THE LESSON

Any questions?

EE4047 – High Voltage Techniques I

Continuation to:

Breakdown in Solids

Week 14.2 – Thursday Lesson





Asst. Prof. Dr. M. Alparslan

08.06.2023

OUTLINE

1 Breakdown of solid insulating materials

2 Losses in solid dielectrics

3 Breakdown types in solid dielectrics

1 BREAKDOWN OF SOLID INSULATION MATERIALS

Solid dielectrics

- Ceramic
- Glass
- Silicones
- Paper, Oiled paper
- Resin

1 BREAKDOWN OF SOLID INSULATION MATERIALS

In solid dielectrics, highly purified and free of imperfections, the breakdown strength is high, of the order of 10 MV/cm.

The highest breakdown strength obtained under carefully controlled conditions is known as the ‘’intrinsic strength’’ of the dielectric.

LOSSES IN SOLID DIELECTRICS

Losses in solid dielectrics

- Dielectric losses
- Dipole losses
- Hysteresis losses
- Ionization losses

The common outcome, **material heats up**

3 LOSSES IN SOLID DIELECTRICS

Breakdown types in solid dielectrics

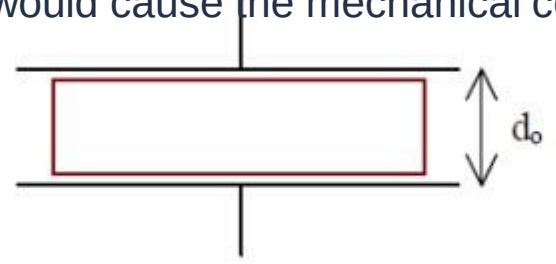
- Instantaneous breakdown
- Electromechanical breakdown
- Electrical breakdown
- Cavity discharge (internal discharge)
- Treeing
- Thermal breakdown

LOSSES IN SOLID DIELECTRICS

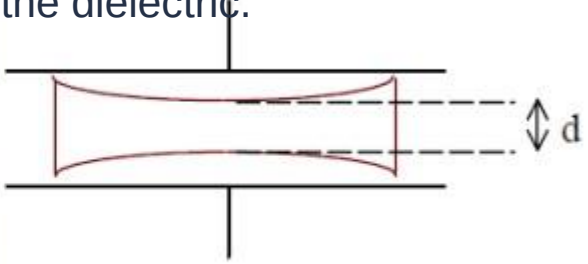
Electromechanical breakdown

When an electric field is applied to a dielectric between two electrodes, a mechanical force will be exertes on the dielectric due to the force of attraction between the surface charges. This can be also due to deformation of the structure of the insulating material by time during operation.
This compression decreases the dielectric thickness, thus increasing effective stress.

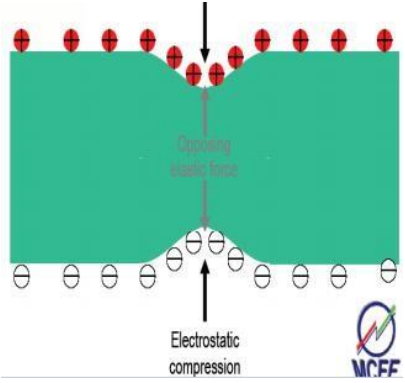
When the field is increased, the thickness of the material decreases. Increase in the field may would cause the mechanical collapse of the dielectric.



In the absence of field



With applied field

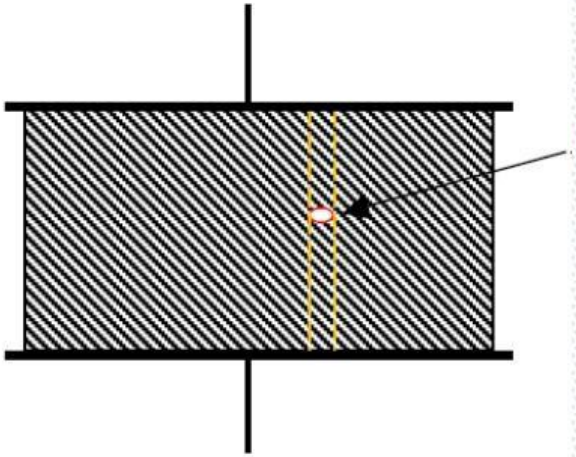


Breakdown due to internal discharges

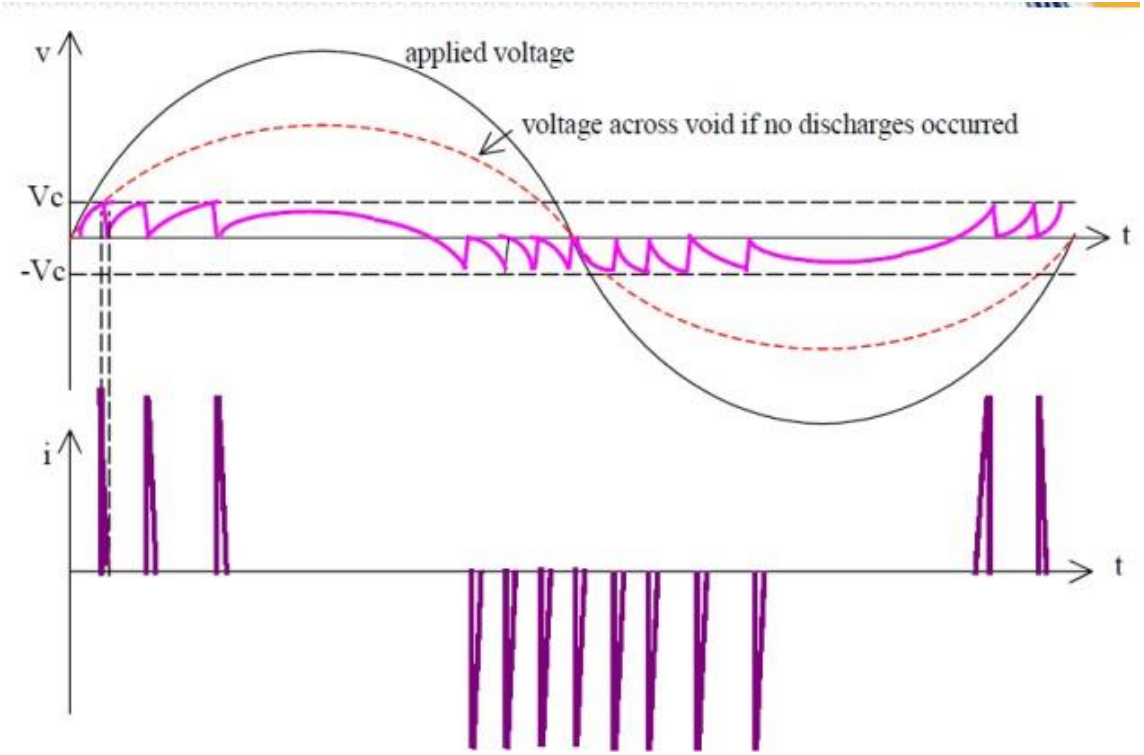
Solid insulating materials sometimes contain voids or cavities in the medium or boundaries between the dielectric and the electrodes.
These voids have a dielectric constant of unity and a lower dielectric strength.

Hence the electric field strength in the voids is higher than that across the dielectric.

Thus even under normal working voltages, the field in the voids may exceed their breakdown value and breakdown may occur.



Breakdown due to internal discharges



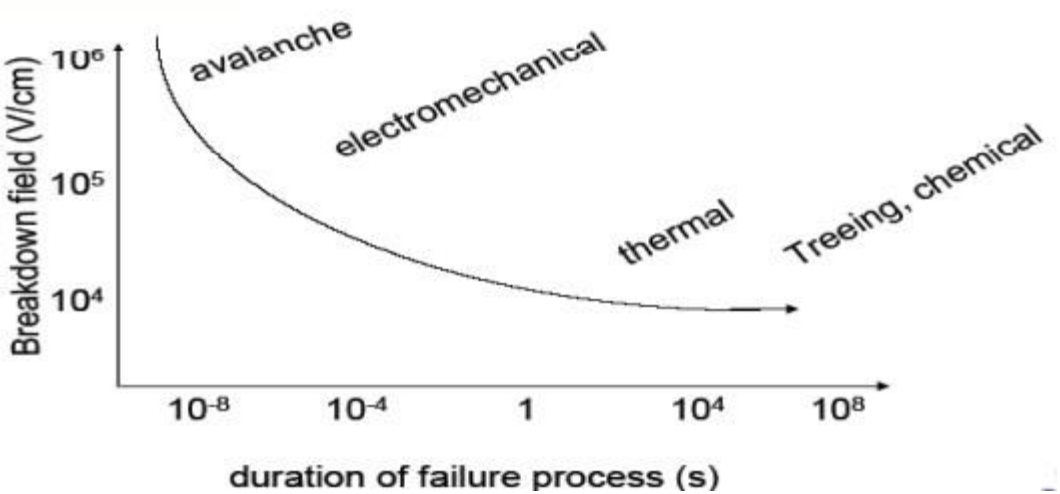
Thermal breakdown

Heat is generated continuously in electricaly stressed insulation by dielectric losses, which is transferred to the surrounding medium by conduction through the solid dielectric and by radiation from its outer surfaces.

If the heat generated exceeds the heat lost to the surroundings, the temperature of the insulation increases.

LOSSES IN SOLID DIELECTRICS

Time and field of failure of different breakdown mechanisms



END OF THE LESSON

Any questions?

Check out the following videos on Youtube:

Breakdown voltage testing of transformer oil: https://www.youtube.com/watch?v=nZAg_A1tb0o

Animation of transformer oil testing: https://www.youtube.com/watch?v=BnrARAP2btk

Transformer oil test on site training: https://www.youtube.com/watch?v=YTp-SjvaA5Q

Oil testing equipment video: https://www.youtube.com/watch?v=WxVKADaLyXk

List of Topics

After the Midterm:

- Cylindrical Electrode Systems
- Electrical Breakdown in Gases
- Breakdown in Liquids
- Breakdown in Solids
- Corona Discharge
- Refraction of Electric Fields
- Approximated Electric Field Calculations
- Finite Difference Method
- Group Project Presentations on HV Test and HV Laboratories
- Academic and Industrial Invited Expert Speaker Lesson

EE4047 – High Voltage Techniques I

Study Questions:

Invited Expert Speaker Lesson

Asst. Prof. Dr. M. Alparslan Zehir

STUDY Questions

1. Could you briefly explain ITU Fuat Kulun High Voltage and your major activites in that facility?
2. Among the subfields and topics of profession in High Voltage, what do you speficially focus on?
3. How would you decribe the field of High Voltage, compared to the other professions in Electrical and Electronics Engineering? (in terms of difficulty, required background, career opportunities, technological developments and other)
4. What are the career opportunities in the area of High Voltage? How would you compare it with other areas of Electrical and Electronics Engineering?
5. What are the working conditions and provided availabilities in the area of High Voltage? How would you compare it with other areas of Electrical and Electronics Engineering? (salary, working environment, future career pathways and other)
6. Which skills are required in High Voltage industry and academia? (softwares, mathematical and physical background, practice, hardware and hardware in the loop knowledge and other)
7. What are the major public and private organizations about High Voltage in Turkey?
8. What are the major public and private organizations about High Voltage in Europe?
9. What are the major public and private organizations about High Voltage in the World?
10. What are the major universities about High Voltage in Turkey?
11. What are the major universities about High Voltage in Europe?
12. What are the major universities about High Voltage in the World?
13. What are the current and near future challenges, issues, problems about High Voltage in the industry and academia?
14. What are the current and near future trends about High Voltage in the industry and academia?
15. How are the academic opportunities about High Voltage, compared to other areas of Electrical and Electronics Engineering?
16. What are the difficulties, challenges or disadvantages of the area of High Voltage, in industry and academia?

EE4047 – High Voltage Techniques I

Study Questions:

Cylindrical Electrode Systems

Asst. Prof. Dr. M. Alparslan Zehir

STUDY Questions

1. Which equipment and components in HV systems can be considered as cylindrical electrode?
2. What is a coaxial cylindrical electrode system? Draw an example.
3. In the voltage formula, the minimum electric field strength formula and the maximum electric field strength formula for coaxial cylindrical electrode systems, which symbol represents which parameter in the real system?
4. Draw equipotential lines and field lines in an example coaxial cylindrical electrode system.
5. How a multilayer coaxial cylindrical electrode system can be represented, using capacitors?
6. Consider a multilayer concentric spherical electrode system numerical example we solved in previous weeks as a coaxial cylendrical electrode system. Using the same given numerical values, but using the formulas for cylindrical electrode systems, please solve the example.
7. How uniform stress condition can be ensured in multilayer coaxial cylindrical electrode systems?
8. In actual electrode seperation formula, effective electrode seperation formula, geometric characteristic formula and utilization factor (electrode efficiency formula), which symbol represents which parameter in the field?

EE4047 – High Voltage Techniques I

Study Questions:

Electrical Breakdown in Gases

Asst. Prof. Dr. M. Alparslan Zehir

STUDY Questions

1 Explain Townsend Theorem in details. Also explain its two coefficients. Draw figure if needed.

2 Explain streamer theory. Draw figure if needed.

3Show electrode gap-breakdown voltage for three different electrode geometry and polarities on a single graph.

4 Explain breakdown, flashover and partial discharge. What is the difference between each?

5 Explain ionization by collision in details.

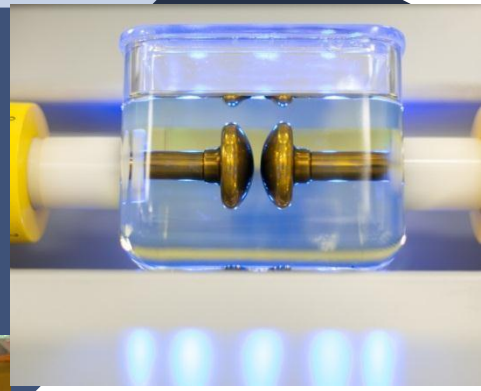
6 Explain photo-ionization in details.

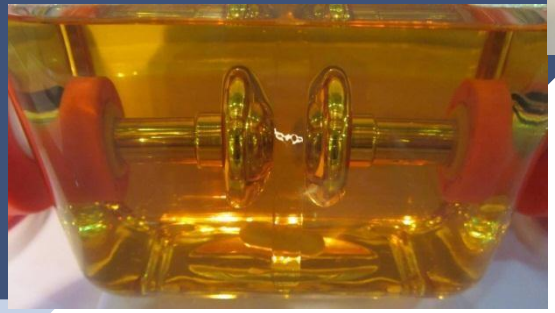
7 Explain thermal ionization in details.

8 Explain surface ionization in details, with its four sub categories.

EE4047 – High Voltage Techniques I

Study Questions for:
Breakdown in Liquids





Asst. Prof. Dr. M. Alparslan Zehir

Study Questions

1. What are the three commonly used liquid dielectrics?
2. What are the three main reasons for using liquid dielectrics?
3. Write the 8 factors that affect breakdown strength and breakdown voltage in liquids. Briefly describe how they affect breakdown strength and breakdown voltage?
4. Explain the three types of impurities and how they behave in liquid dielectrics.
5. Write the three types of breakdown in liquid dielectrics and briefly describe each.
6. How dissolved gasses , water globules and solid particles behave in liquid dielectrics? Describe each seperately.
7. Write the desired dielectric oil properties.
8. On a $U_{breakdown}/electrodegap$ graph, show the impact of electrode polarity and geometry on breakdown voltage.

EE4047 – High Voltage Techniques I

Study Questions:

Breakdown in Solids

Asst. Prof. Dr. M. Alparslan Zehir

STUDY Questions

1 List three solid dielectric materials.

2 Write the names of four losses in solid dielectrics.

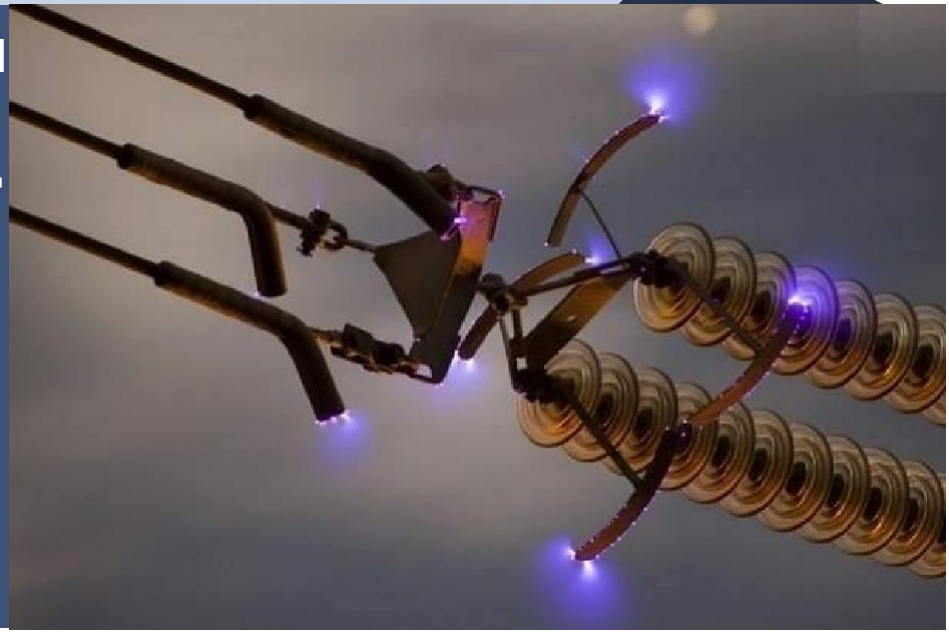
3 List three breakdown types in solid dielectrics.

4What is electromechanical breakdown? How it happens?

5- What is thermal breakdown? How it happens?

EE4047 – High Voltage Techniques I

Study Questions for
Corona Discharge



Asst. Prof. Dr. M. Alparslan Zehir

Study Questions for Corona Discharge

1. What is the literal meaning of corona?

2.What is the physical meaning of corona in HV? Which places-parts of equipment does it occur?

3.What are the three main indicators of corona discharges? What are the impacts of these indicators?

4. How weather conditions impact corona?

5. In the corona voltage formula, please describe which symbol represents which parameter in the real world.

6. In the power loss formula (Peek's formula) please describe which symbol represents which parameter in the real world.

7. How the corona discharges can be minimized?

8. Explain the ozone gas generation related problems of corona discharges.

9. Explain the corona discharge problems in SF_6 gas insulated systems.

EE4047 – High Voltage Techniques I

Study Questions:

Refraction of Electric Fields

Asst. Prof. Dr. M. Alparslan Zehir

STUDY Questions

1 In which cases we focus on refraction of electric fields?

2 Show the refraction of electric field lines and equipotential lines on an example case, labelling the angles, and the vectors.

3Solve the two examples again, using different parameters, modifying the scenarios given in the question.

EE4047 – High Voltage Techniques I

Study Questions:

Approximated Electric Field Calculations

Asst. Prof. Dr. M. Alparslan Zehir

STUDY Questions

1 A plane-sphere electrode system can be approximated to which system? How the original equation is modified?

2A sphere-sphere electrode system can be approximated to which system? How the original equation is modified?

3Solve the sphere-sphere numerical example again, using different parameters, modifying the scenario given in the question.

4A plane-cylinder electrode system can be approximated to which system? How the original equation is modified?

5 A cylinder-cylinder electrode system can be approximated to which system? How the original equation is modified?

6 Solve the cylinder-cylinder numerical example again, using different parameters, modifying the scenario given in the question.

EE4047 – High Voltage Techniques I

Study Questions:

Finite Difference Method

Asst. Prof. Dr. M. Alparslan Zehir

STUDY Questions

1 What is the general formula for calculating the voltage of a selected point, using finite difference method.

2 Solve the two examples again, using different parameters, modifying the scenarios given in the question.

EE4047 – High Voltage Techniques I

Study Questions:

Group Projects on HV Test and HV Laboratories

Asst. Prof. Dr. M. Alparslan Zehir

STUDY Questions

1 List the names, cities and countries of three HV laboratories presented by the groups.

2 List the names of three tests presented by the groups. Explain a test presented by another group in details (not your own group’s test).

END OF THE LESSON

Any questions?